# TSL3000 Robot Controller

# Controller Manual

(Compatible Robots: THL/THE Series SCARA Robot)

Rev.7
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Important

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Shibaura Machine

This instruction manual applies to the controller TSL3000.
The compatible robots of the TSL3000 are the THL/THE series robots.

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# Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the standard specifications, installation, wiring, and maintenance of the controller. Read this manual before unpacking the product.
Read also"ST85358-8_THL Safety Manual" or "STA1112-2_THE_Safety Manual" to understand the safety measures for using the controller.

This manual has the following chapters.

Chapter 1: Specifications
This chapter describes the standard specifications of the controller.

Chapters 2 to 4: Installation
These chapters describe transport and installation of the controller, and precautions for temporary storing the unpacked controller.
These chapters also describe the installation environment, required space, and installation procedure for the controller, and wiring between the controller and other controllers or peripheral devices.

Chapters 5 to 7: Maintenance
These chapters describe maintenance of the controller.

Chapter 8: Disposal
This section describes the disposal of controllers.

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## Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer's responsibility.

# Use Method

The use conditions for the THL/THE series robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.3 **Controller Specifications**", Installation "3.1 **Installation Environment**", and "THL/THE Series Robot Manual Specification Part" in the Instruction manual.

2. Restrictions on Use: Tables 1 and 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1　Restricted Specification Sheet (1)

| 1. Purpose and application of using product (restrictions on use) | |
|---|---|
| 1) Intended use | Allow the robot to achieve operation along with the purposes (article conveyance, assembly, inspection, machining, etc.).<br>(1) During teaching and adjustment operation<br>- Operation is likely to be performed by turning on the servo power within the movable range of the robot.<br>- A person who observes teaching and operation checking worker as a co-worker is arranged outside the movable range.<br>(1) Teach the position where the robot operates.<br>(2) Create the operation sequence (program, etc.).<br>(3) Transfer and check signal and data with the external equipment.<br>(4) Check operation at low speed (250 mm/sec or less).<br><br>(2) During operation<br>- Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) are taken to keep a person away from the operating robot.<br>- When the operating robot is approached, measures shall be taken to turn off the servo power.<br>(1) Start the robot.<br>(2) Perform operation along with the purpose at the specified speed.<br>(3) Stop the robot.<br><br>(3) During maintenance and inspection<br>- Contents performed within the movable range of the robot when the primary power is off<br>(1) Check the bolts for looseness and backlash and re-tighten them.<br>(2) Check the cable clamp for looseness.<br>(3) Inspect damage.<br>(4) Clean dust.<br>(5) Inspect and apply antirust agent.<br>(6) Check the timing belt for looseness and crack.<br>(7) Check the cable and air tube for wear.<br>(8) Check the operation when each axis is pushed and moved by hand.<br>(9) Replace the consumable parts for the robot and controller.<br>(10) Replace the battery for memory in the controller.<br><br>- Contents performed within the movable range of the robot when the primary power is on<br>(1) Check the grease state and apply grease (when the servo power is off).<br>(2) Push each axis by hand and check it for backlash (when the servo power is on).<br>(3) Check abnormal vibration and sound during operation (when the servo power is on).<br>(4) Check the teaching point for misalignment (when the servo power is on).<br>(5) Replace the battery for robot position detector (when the servo power is off). |

Table 2　Restricted Specification Sheet (2)

| | | |
|---|---|---|
| 2) Predictable misuse | - Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot<br>- Modification of robot body, controller, and optional parts and energization with these covers removed<br>- Operation in other than the specified installation state (floor, suspension, and wall-mounted)<br>- Operation on the frame that does not satisfy the specified strength<br>- Operation in other than the specified environment | |
| 3) Unexpected start | - Malfunction caused by excessive noise<br>- Robot start caused by start signal unexpectedly sent from the peripheral equipment to the robot<br>- Unintended start caused by abnormal communication data<br>- Malfunction caused by voltage fluctuation | |
| 2. Replacement of product components (restrictions on time) | | |
| 1) Mechanical restriction<br>2) Electrical restriction | Replace consumable parts as required or periodically through daily inspection, regular inspection (every 3 and 6 months and every year) and overhaul (every 5 years). | |
| 3 Movable range of product (restriction on space) | | |
| 1) Operating range | Axis 1 | Conforms to "THL/THE Series Robot Manual Specification Part" in the Instraction manual. |
| | Axis 2 | Conforms to "THL/THE Series Robot Manual Specification Part" in the Instraction manual. |
| | Axis 3 | Conforms to "THL/THE Series Robot Manual Specification Part" in the Instraction manual. |
| | Axis 4 | Conforms to "THL/THE Series Robot Manual Specification Part" in the Instraction manual. |
| | Additional axis | Conforms to individual specifications. |
| | — | — |
| | — | — |
| 2) Interface | - The product shall be installed at a location suitable for outer shape and operating range.<br>- Space required for maintenance and inspection shall be secured. | |
| 3) Work environment | - Danger prevention measures that prevent contact with the robot (installation of the fence or enclosure) shall be taken.<br>- Space shall be secured to prevent a person from being caught between the robot and the fence or AY15enclosure. | |
| 4. Life cycle of product | | |
| During the life cycle of assembly, shipment test, transportation, installation, teaching, operation, maintenance and disposal, risk assessment is carried out for “installation” stage (transportation and installation), “use” stage (teaching, operation, and maintenance) and disposal. | | |
| 5. Person who approaches the machine | | |
| Target | Knowledge, experience, and condition | |
| 1) Transportation and install | Knowledge: Person who has understood the instruction manual | |
| | Experience:— | |
| | Condition: Qualified person on forklift or crane (as required) | |
| 2) Teaching/adjustment operation worker (including co-worker) | Knowledge: Person who has understood the instruction manual | |
| | Experience: Person who has acquired the operation of the robot to be used | |
| | Condition: Person who has received special training pertaining to the operation such as teaching of the industrial robot and to whom work has been permitted by the employer | |
| 3) Operation worker (operator) | Knowledge: Person who has understood the instruction manual | |
| | Experience: Person who has understood the work regulation | |
| | Condition:— | |
| 4) Maintenance and inspection worker | Knowledge: Person who has understood the instruction manual | |
| | Experience: Person who has acquired the operation of the robot to be used and person who has acquired maintenance work of the target robot | |
| | Condition: Person who has received special training pertaining to the operation such as inspection of the industrial robot and to whom work has been permitted by the employer | |
| 5) Disposal worker | Knowledge: Person who has understood the instruction manual and the disposal method for each local government | |
| | Experience:— | |
| | Condition:— | |
| 6. Other | | |
| | | |

## Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.

We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period

The warranty period of the product is either of the following, whichever is earlier.

1) 18 months after delivery from our factory
2) 12 months after installation at your site

2. Warranty scope

1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
   Any repairing after the warranty period passes will be charged.

3. Disclaimer

The warranty shall not apply to the following cases.

1) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
2) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables[*1]).
3) Any failure regarding sensuous phenomena such as noise not affecting the functions
4) When any modification or disassembly without our consent is confirmed
5) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
6) Any failure or damage due to act of God, fires, and other force majeure
7) Internal data such as programs and points created or modified by users
8) When a product purchased in Japan is taken abroad

4. Precautions

1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
2) If the customer did not observe the "WARNING" and "CAUTION" described in this instruction manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.
3) Note that "WARNING" and "CAUTION", and other information are those assumed by SHIBAURA MACHINE.

(*1) Consumable parts refer to the replacement parts for maintenance as listed in Chapter 7 in this instruction manual.

## Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy
Step 2: Safety protection - Installation of protective fences, etc.
Additional protection measures - Installation of emergency stop devices
Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure. Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

●ISO 10218, ISO 10218-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

The following describes the major Japanese standards for industrial robots.

Safety standards for industrial robots are defined by the Japanese Industrial Standards (JIS) by applying ISO 10218 and ISO 10218-1 without changing any of their technical contents and configuration.

●JIS B 8433, JIS B 8433-1

Japanese laws and regulations for safety of industrial robots are defined as follows.

● Industrial Safety and Health Act, Article 59 (Safety and Health Education)

"The employer shall, when a worker is to be placed in the dangerous or harmful operations, give the worker the special education for safety and/or health concerning the said operations."

Operations requiring special education for handling industrial robots are defined as follows.

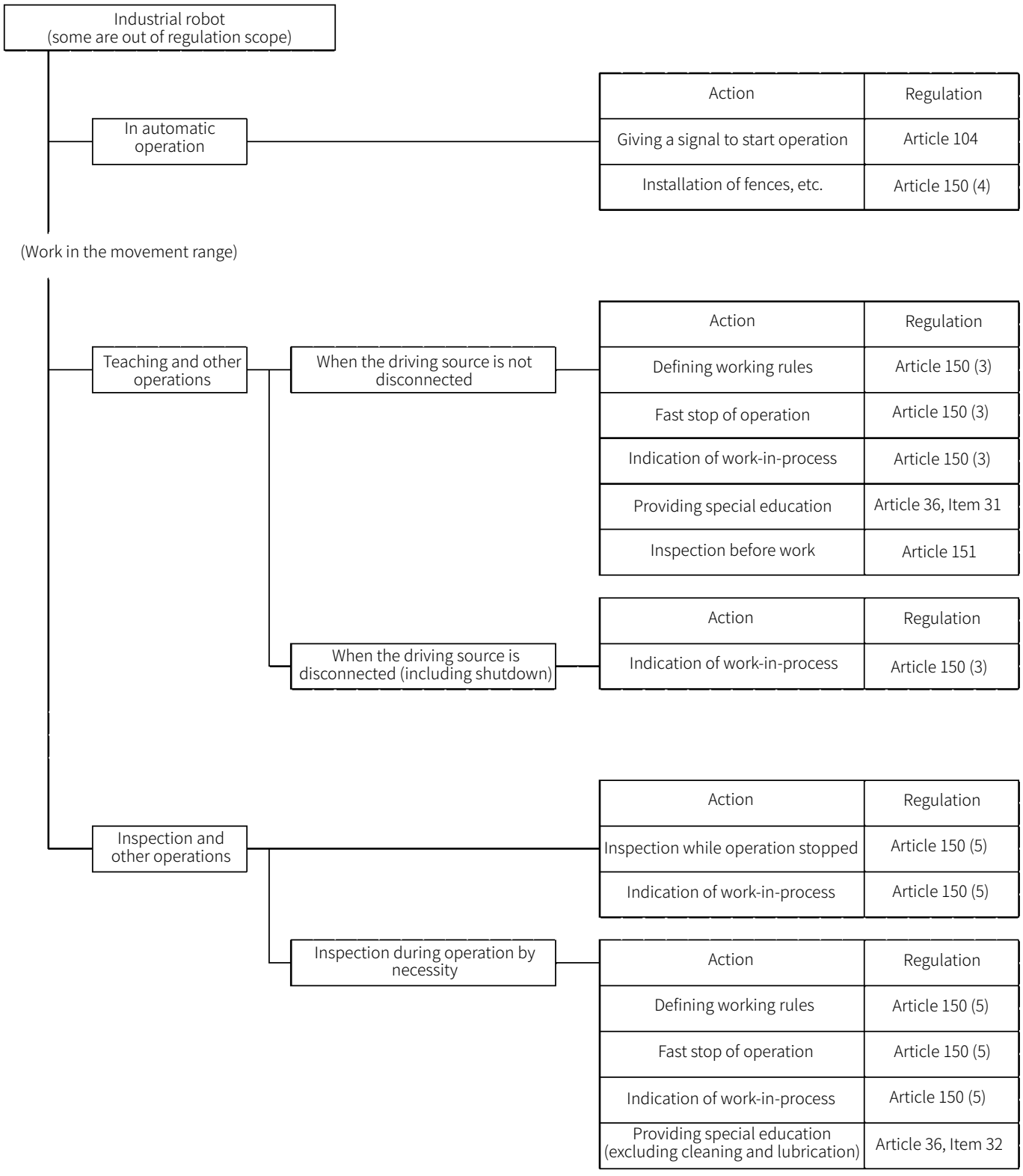
● Labor Safety and Health Regulation, Article 36 (Operations Requiring Special Education)

• Item 31, Teaching and other operations of the industrial robots
• Item 32, Inspection and other operations of industrial robots

Measures that should be taken to prevent danger in operations handling industrial robots are defined as follows.

● Labor Safety and Health Regulation, Section 9 (Industrial Robot)

• Article 150 (3), Teaching, etc.
• Article 150 (4), Preventing danger during operation
• Article 150 (5), Inspection, etc.
• Article 151, Inspection




System diagram of the Labor Safety and Health Regulation

## Safety Precautions

This manual contains important information for the safe use of the equipment and the prevention of injury to the operator and others and the prevention of damage to property.
To ensure safety from installation to operation of the controller, be sure to read the separate instruction manuals "ST85358-8_THL Safety Manual" or "STA1112-2_THE_Safety Manual" carefully before actually starting work.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

### [Meanings of Indications]

| Indication | Meaning |
|---|---|
| ⚠DANGER | Indicates the imminent danger that incorrect operation could cause death or serious injury. |
| ⚠WARNING | Indicates the danger that incorrect operation could cause death or serious injury. |
| ⚠CAUTION | Indicates that incorrect operation could cause injuries[1)]. |
| NOTICE | Indicates the possibility that only damage to property[2)] may occur. |

*1: Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
*2: Damage to property indicates extensive damage related to the destruction of assets or materials.

### [Meanings of symbols]

| Symbol | Meaning |
|---|---|
|  | Indicates a prohibited action.<br>The specific content of the prohibition is shown pictorially or in text form inside the symbol or nearby it. |
|  | Indicates an instructed (mandatory) action.<br>The specific content of the instruction is shown pictorially or in text form inside the symbol or nearby it. |
|  | Indicates a precaution or warning (things to which attention should be paid).<br>The specific content of the precaution is shown pictorially or in text form inside the symbol or nearby it. |

# Specifications

This part describes the basic specifications and the names of components of the TSL3000.

# 1 Specifications

## 1.1 Equipment Configuration

Figure 1.1 shows the robot equipment configuration.
This controller supports the THL Series (THL300 to 1200) and THE Series (THE400) SCARA robots.
For connecting the 200V power supply, refer to 4.1.2 "ACIN" connector for connecting the power cable (Figure 4.1-①). For connecting the 24V power supply, refer to 4.1.7 "GNDP24V" connector for connecting the I/O signal power cable (Figure 4.1-③).

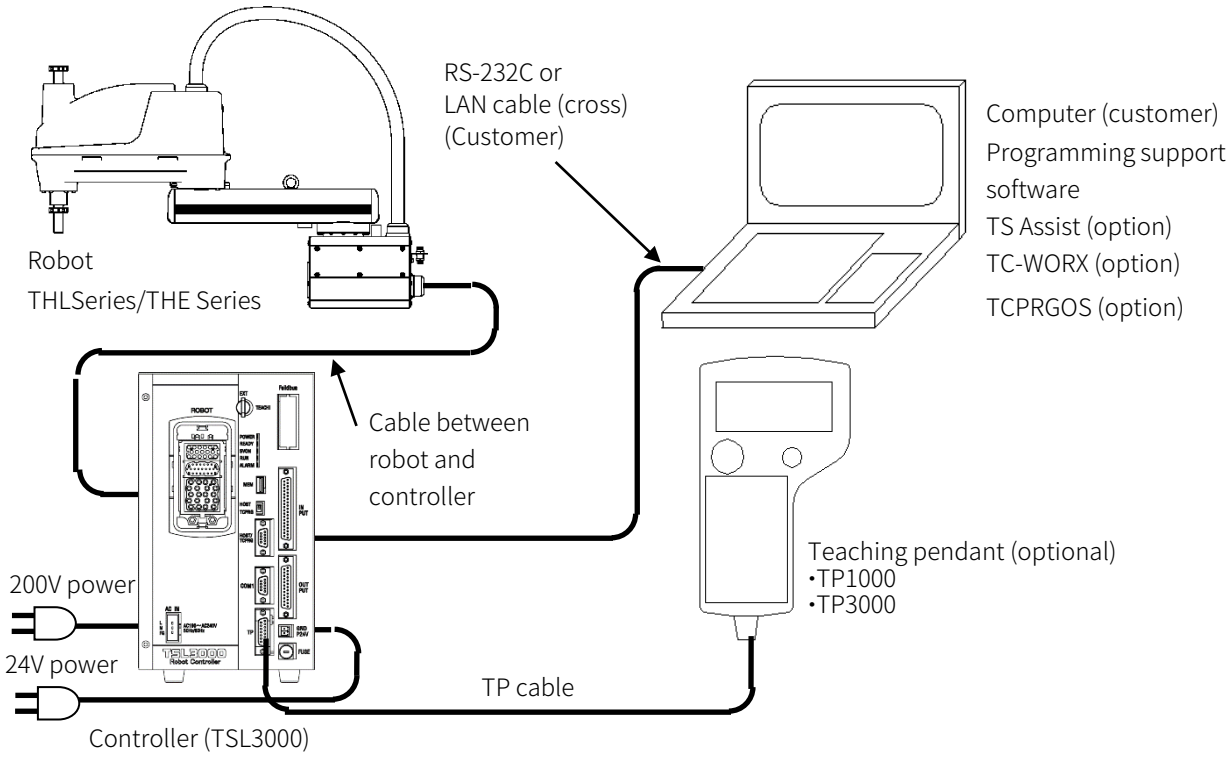



Figure 1.1 Robot equipment configuration

## 1.2 Names of Components

Figure 1.2 shows the names of components.

For connecting the ACIN, ROBOT and GNDP24V connectors, refer to Chapter 4.

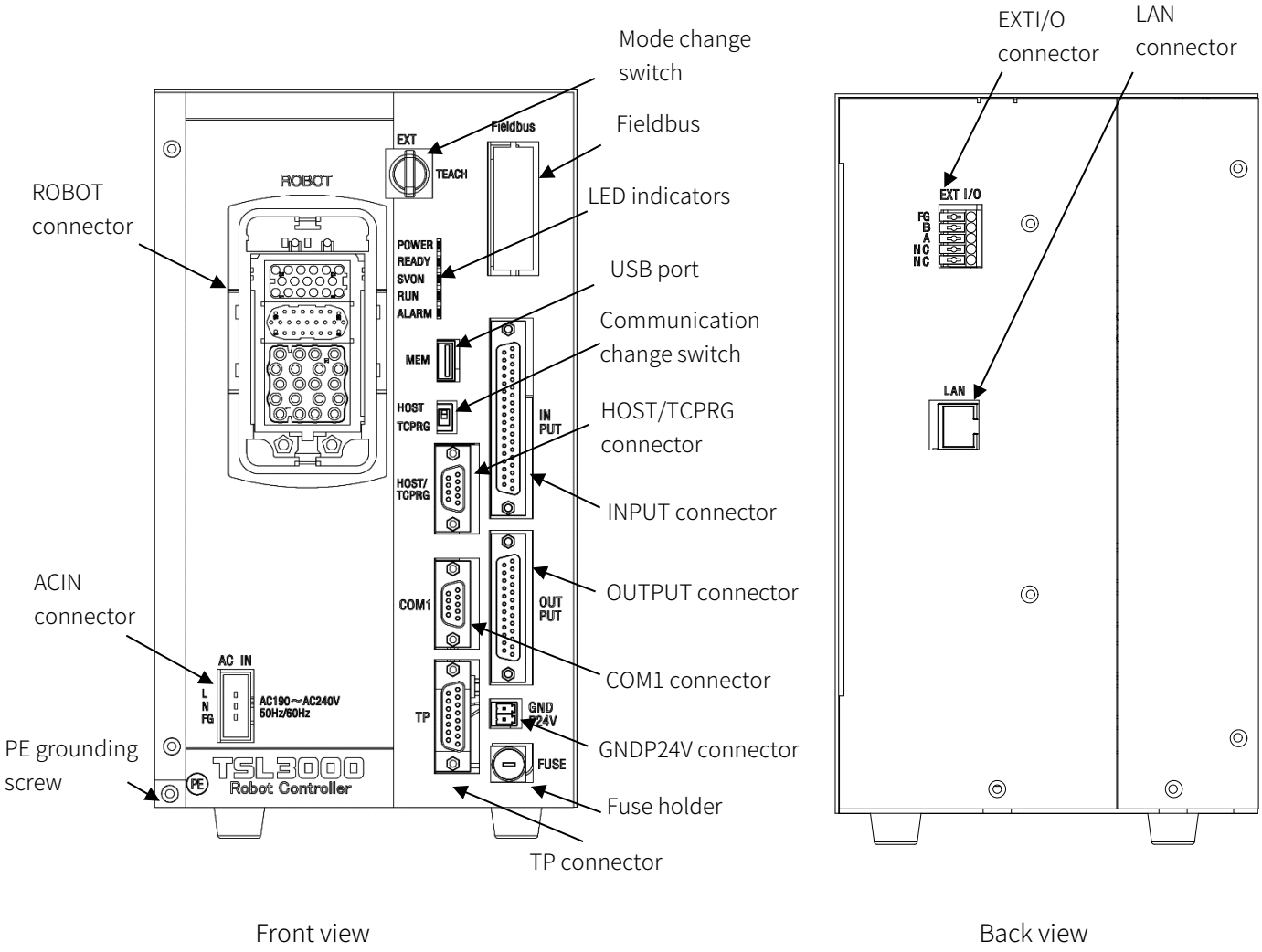



Figure 1.2 Names of components

## 1.3 Controller Specifications

| No. | Item | | Specifications | Remarks |
|---|---|---|---|---|
| 1 | Number of controlled axes | | Simultaneous 4 axes maximum | |
| 2 | Operating mode | | PTP, CP (line, arc), Short cut, arch operation | |
| 3 | Servo type | | Digital servo | |
| 4 | Storage capacity | | Total: Approx. 6400 points + 12800 steps<br>1 program: Approx. 2000 points + 3000 steps | 0.5 Mbytes |
| 5 | Number of programs that can be registered | | Up to 256 (247 user files and 9 system files) | |
| 6 | Auxiliary memory | | USB memory | |
| 7 | Storage type | | Battery backup RAM | |
| 8 | Position detection method | | Absolute encoder | |
| 9 | Teaching method | Point teaching | Remote: Control via the teaching pendant | |
| | | | Coordinates: Input the X, Y, Z, C, and T values via the teaching pendant.<br>Servo free: Arm movement via input | |
| | | Program input | Input via the teaching pendant | |
| 10 | External I/O signals | | 8 inputs, 8 outputs | |
| 11 | Band control signals | | Determined by the robot specifications | |
| 12 | External control signals | Input | Program select, start-up, stop, program reset, etc. | |
| | | Output | Servo ON, operation ready, failure, cycle stop, etc. | |
| 13 | Serial communication port | | RS-232C: 1 port (HOST/TCPRG), dedicated type<br>RS-232C: 1 port (COM1) general-purpose<br>RS-422: 1 port, TP1000, dedicated type | HOST/TCPRG is supported by the change switch. |
| 14 | Speed setting | | Override/Speed limit/Program instructions, 1 to 100% for each instruction | |
| 15 | Acceleration setting | | Program instruction: 1 to 100% | |
| 16 | Torque Control | | Program instruction: 1 to 300% | |
| 17 | Teaching device | | Teaching pendant | |
| 18 | Coordinate system | | The Base, Workpiece, and Tool coordinates can be set. | |
| 19 | Movement limit | | Soft limit | |

| No. | Item | | Specifications | Remarks |
|---|---|---|---|---|
| 20 | Self-diagnostic function | | Error detection, etc. | |
| 21 | Interrupt function | | Starts an interrupt program via an input signal, timer, etc. | |
| 22 | Operating mode | | TEACH mode, External Auto mode (SIGNAL), External Auto mode(RS232C)/ External Auto mode(ETHERNET) | |
| 23 | Operating method | Internal | Continuous, Cycle, Step, Operation step | |
| 24 | | External | Cycle, Continuous | |
| 25 | Controller main unit | Dimensions | 150(W) × 266(H) × 304(D) | The height includes the rubber foot height. |
| | | Weight | 7 kg | |
| | | Color | Black | |
| | Power supply | | Single phase 190 to 240 VAC, 50/60 Hz | |
| 26 | Computer software TSPC | | Program creation, teaching, remote operation, etc. | |
| 27 | Programming language | | SCOL | |

[Specifications of Controller Options]

| No. | Name of product | Specification | Remarks |
|---|---|---|---|
| 1 | Teaching pendant | TP1000 | Standard cable: 5 m |
| 2 | | TP3000 | Standard cable: 5 m |
| 3 | Teaching pendant cable extension | | 10m/15m |
| 4 | Controller Output Type P spec. | | +Common |
| 5 | External input signal cable | | Cable length: 6 m |
| 6 | External output signal cable | | Cable length: 6 m |
| 7 | Controller mounting bracket | Mounting Bracket | 2 pieces/set |
| 8 | Extended I/O expansion | TR48DIOCN-1 | Type-N |
| | | TR48DIOC-1 | Type-P |
| 9 | Program support software | TSAssist | Instruction Manual supplied |
| 10 | TCmini (built-in PLC) editing software | TC-WORX | Can be used with Windows10 |
| 11 | | TCPRGOS | Cannot be used with Windows10 |
| 12 | Field bus extensions | PROFIBUS | *1 |
| 13 | | DeviceNet | |
| 14 | | CC-Link | |
| 15 | | EtherNet/IP | |
| 16 | | EtherCAT | |
| 17 | | PROFINET | |

*1 : Select either PROFIBUS, DeviceNet, CC-Link, EtherNet/IP, EtherCAT, or PROFINET for the slave module for the field bus. Customers need to decide which field bus they use when placing an order. PROFIBUS is a trademark of Profibus User Organization. DeviceNet is a trademark of ODVA. CC-Link is a registered trademark of Mitsubishi Electric. EtherNet/IP is a trademark of ODVA. EtherCAT is a registered trademark of Beckhoff Automation GmbH.
For details of the field bus, refer to the ST80756 TS3000 Instruction Manual for Field Bus Slave.

# Installation

This part describes transport, unpacking, carrying, and installation procedures of the controller.

# 2 Transport, Unpacking, and Carrying

The transport of the controller can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

## 2.1 Transporting the Controller

| ⚠CAUTION | |
|---|---|
|  <br>**Prohibited** | ● Do not stack package boxes. Doing so may result in collapse and injury. |
|  <br>**Instruction** | ● Wear protective gloves and protective shoes when you transport the controller by hand. Failure to do so may result in pinching of your fingers or feet.<br>● Use a dolly or similar device for transport of the controller. Transporting by hand may cause the controller to fall and your fingers and feet to be caught. |

**NOTICE**

**Prohibited**

- Do not expose the controller to excessive vibration during transport. Doing so may cause damage to the controller.
- Do not transport the controller upside down or store it on its side. Doing so will result in damage to the controller.
- Do not allow the controller to get wet. Doing so will cause moisture to enter the interior of the controller and result in malfunction.

## 2.2 Unpacking the Controller

The controller is shipped in corrugated cardboards.
Figure 2.1 shows the packing state. Open the packages in a location easily assessable, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport. For the controller accessories, see the accessory list packed with the controller.

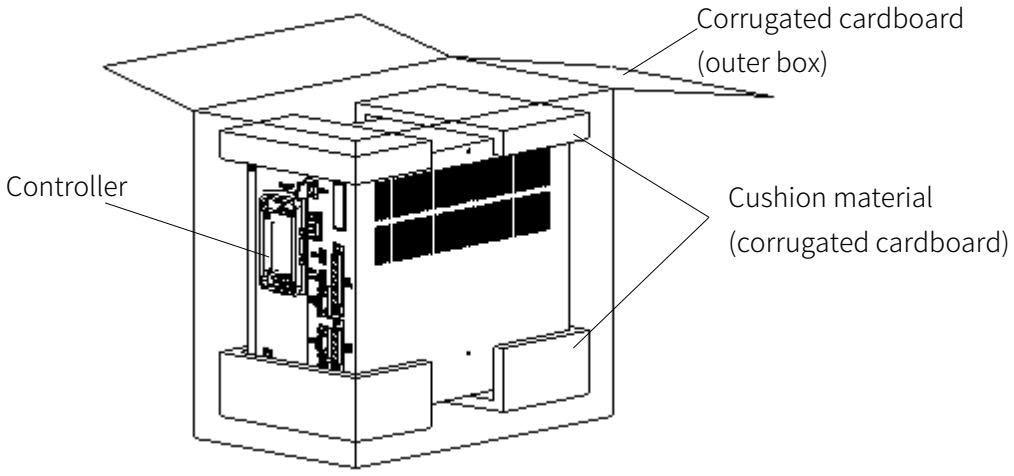



Figure 2.1 Controller packing state

**⚠CAUTION**

**Instruction**

- Wear protective equipment when disposing of unneeded wooden pallets or cardboard boxes after unpacking. Sharp edges or protruding nails can sting and cause injury.
- Use a cardboard cutter or similar tool when unpacking boxes. Use of sharp-edged cutters may result in cuts to your fingers.
- Perform unpacking on a stable surface. Unpacking on an unstable surface can cause the controller to fall and your hand or foot to be caught, and also cause back injury.
- Wear protective gloves when unpacking the controller. Failure to do so may result in injury if your hand gets caught between the cardboard and the controller, and also result in injury from the corners of the controller and the cardboard.

| NOTICE | |
|---|---|
|  **Prohibited** | ● Do not allow the controller to get wet. Doing so will cause moisture to enter the interior of the controller and result in malfunction. |

## 2.3 Carrying the Controller

Disconnect and remove cables and teaching pendant from the controller before transport.
Do not apply an excessive force to the controller. Otherwise the controller may be damaged.

| ⚠CAUTION | |
|---|---|
|  **Prohibited** | ● Do not carry the controller by grabbing the AC power supply unit. Doing so may cause breakage, falling of the controller, and pinching of your feet, and also may result in an electric shock if AC power is applied to the controller. |
|  **Instruction** | ● Wear protective gloves and protective shoes when you transport the controller by hand. Failure to do so may result in pinching of your fingers or feet.<br>● Use a dolly or similar device for transport or carrying of the controller. Carrying by hand may cause the controller to fall and your feet to be caught.<br>● Carry the controller with the cables disconnected. Transporting the controller with the cables connected may result in wire breakage and electric shock. |

| NOTICE | |
|---|---|
|  **Prohibited** | ● Do not expose the controller to excessive vibration during transport. Doing so may cause damage to the controller.<br>● Do not allow the controller to get wet. Doing so will cause moisture to enter the interior of the controller and result in malfunction. |

## 2.4 Storage

Avoid storing the controller not being used for an extended period after it is unpacked.
If such long-term storage is not avoidable, follow the precautions below.

### 2.4.1 Precautions for storing the controller

| NOTICE | |
|---|---|
|  **Instruction** | ● Store the controller avoiding direct sunlight.<br>It may result in fault due to heating inside the controller.<br>● Seal the controller in a vinyl bag with desiccant inside to prevent rust and dust. |

# 3 Installation

The installation of the controller can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

**⚠ DANGER**



**Prohibited**

- Never install the controller in a location where it will be exposed to water harsher than its ingress protection rating. Doing so may cause a fire and also an electric shock.
- Never place combustible materials near the controller. Doing so can cause them to catch fire in case of a controller failure that leads to ignition.
- Do not place any object on the controller. Do not stack controllers.
  - Doing so prevents ventilation space and causes heat buildup, which leads to fire.
  - Doing so can lead to collapse of the controller, which causes it to break or your hand or foot to be pinched because it is not designed to allow objects to be placed on its top and cannot endure such loads.



**Instruction**

- Be sure to install the controller in a location which is outside the hazardous area and from which you can look over the robot's operation. If you use the controller in a location which is inside the hazardous area or from which you cannot look over the robot and peripheral equipment, you or workers within the hazardous area may be struck by the robot or peripheral equipment.

| | |
|---|---|
| | ● If it is unavoidable to install the controller inside the hazardous area, keep the teach pendant outside the hazardous area and use the controller's system signals to output the robot's status externally for indication with a signal tower or LED light.<br>If there is no way to check the status of the robot, a worker who has no choice but to check the controller installed in the hazardous area can be struck by the robot or peripheral equipment.<br>● When installing the controller in an enclosed space such as a control panel, ensure air ventilation holes, forced ventilation, or indirect cooling to prevent heat buildup. Installing the controller in an enclosed space can cause heat buildup, resulting in a fire.<br>● Install the controller inside a control panel or in a location where a sufficient distance to the operator can be secured so that the controller cannot come in contact with the operator. If the controller is installed within the moving range of the operator, he/she may trip over the cable and fall, or get an electric shock from contact with the energized parts.<br>● Do not install the controller in an unstable location. Doing so will cause it to fall and your hand or foot to be pinched.<br>● Be sure to perform wiring of the robot after installation.<br>• If you install or carry the robot with its wiring connected, the cables may break, resulting in electric shock.<br>• If you install or carry the robot with its wiring connected, you may be caught by the cable and fall.<br>● Protect the cables to be routed to the control panel with dedicated ducts or other means. If cables are routed without using dedicated ducts, they may break, resulting in a fire or an electric shock.<br>● Install covers or the like on the buttons and switches of the control devices that are connected to the controller. Failure to do so may cause the signal to be turned on by an unexpected contact, resulting in the robot moving, and you may be struck by it.<br>● Provide the operating device with an indication of the controller execution mode. If the execution mode is not known, you may be struck by the robot moving unexpectedly. |

| NOTICE | |
|---|---|
|  **Prohibited** | ● Never apply a current exceeding the rated output. Doing so may cause damage to output elements, circuit boards, etc.<br>● Do not place any object on the controller. Do not stack controllers.<br>• Doing so can lead to collapse of the controller, which causes it to break because it is not designed to allow objects to be placed on its top and cannot endure such loads. |
|  **Instruction** | ● When not using the LAN, OUTPUT, or SD ports, be sure to attach a cap (or dummy SD card for the SD port) to protect them. Failure to do so will result in malfunction due to static electricity.<br>● When installing the controller in an enclosed space such as a control panel, ensure air ventilation holes, forced ventilation, or indirect cooling to prevent heat buildup. Installing the controller in an enclosed space can cause heat buildup, resulting in malfunction. |

## 3.1 Installation Environment

Table 3.1 shows the environment specifications for the controller.

Table 3.1 Environment specifications for the controller

| Item | Specifications |
|---|---|
| Temperature | In operation state: 0 to 40°C<br>In storage state: -10°C to 50°C |
| Humidity | 20 to 80% (no condensation)<br>Do not install the controller in a place where liquid such as water is applied. |
| Altitude | Up to 1000 m |
| Vibration | Do not apply excessive vibration and shock. |
| Dust | Ensure that there is no electric conductive dust.<br>Contact us if you use the controller in a dusty environment. |
| Gas | Ensure that there is no corrosive or flammable gas. |
| Protection degree | IP20 |
| Overvoltage category | IEC60664-1 Class III |
| Electric shock protection | IEC61140 Class I |
| Pollution level | IEC60664-1 Pollution Level 3 |
| Solar radiation | Avoid direct sunlight. |
| Power supply noise | Ensure that there is no surrounding device generating excessive noise. |
| Magnetic field | Ensure that there is no surrounding device generating a strong magnetic field. |
| Surrounding environment | Ensure that there is no iron powder, oil, salt, or organic solvent in the environment. Ensure that water is not applied to the controller. |

## 3.2 Dimensions

Figure 3.1 shows the outline drawing of the controller (THL series). The mass of the controller main unit is 7 kg.

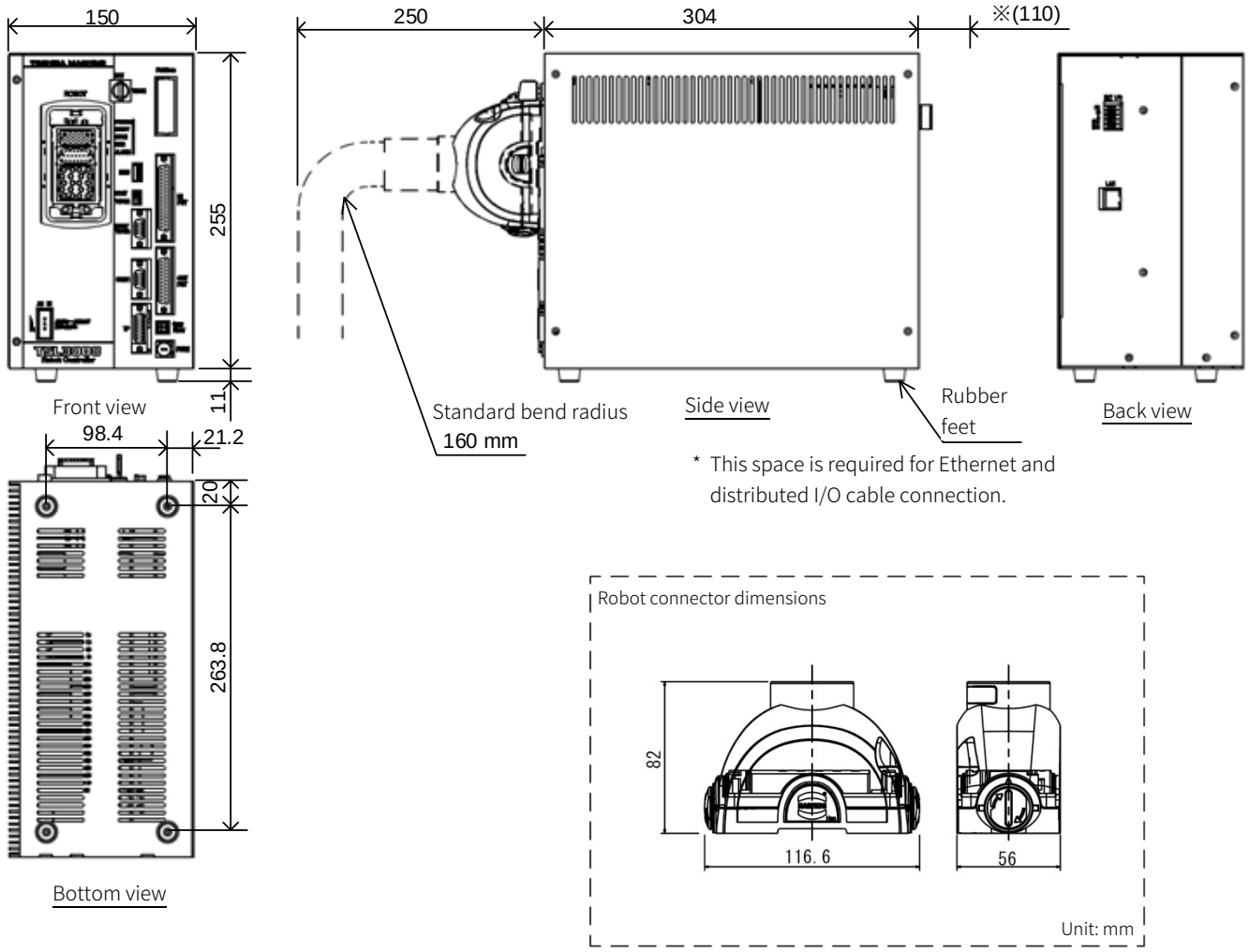



Figure 3.1 Outline drawing of the controller (Robot: THL series)

Figure 3.2 shows the outline drawing of the controller (THE series). The mass of the controller main unit is 7 kg.

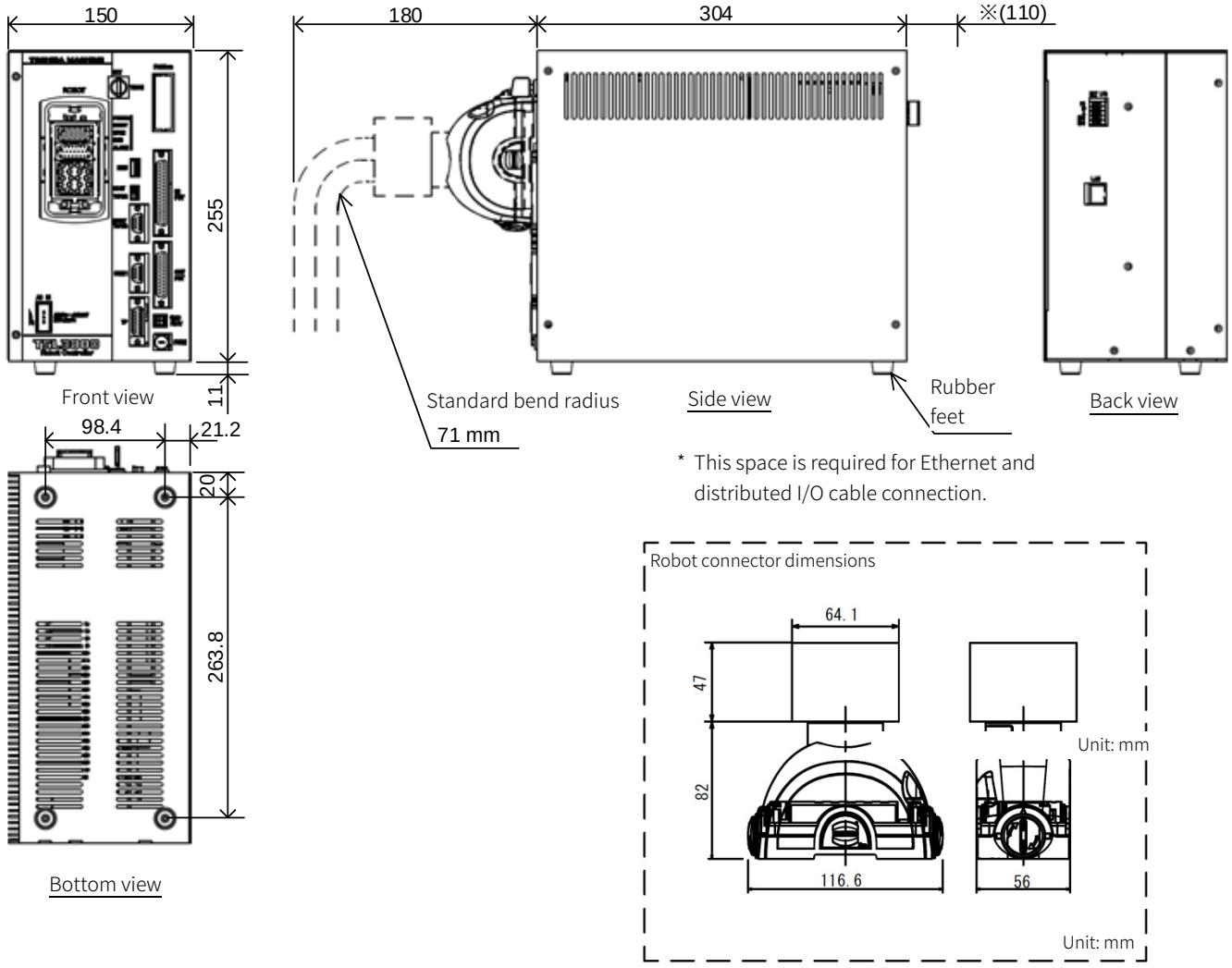



Figure 3.2 Outline drawing of the controller (Robot: THE series)

## 3.3 Precautions for Installation on Floor

Keep 50 mm or more clearance around the sides of the controller and 100 mm or more clearance above the top surface of the controller.

| NOTICE | |
|---|---|
|  **Instruction** | ● Keep ventilation space around the controller at least 50 mm on the right and left sides and back, at least 100 mm on the top, and at least the height of the rubber feet (11 mm) on the bottom. Failure to do so will cause the controller to heat up and malfunction.<br>● When installing the controller in an enclosed space such as a control panel, ensure air ventilation holes, forced ventilation, or indirect cooling to prevent heat buildup. Installing the controller in an enclosed space can cause heat buildup, resulting in malfunction.<br>● When installing the controller inside the control panel, secure it with the fixing bracket or the like. Failure to do so may cause the controller to move inside and fall due to vibration of the control panel. |

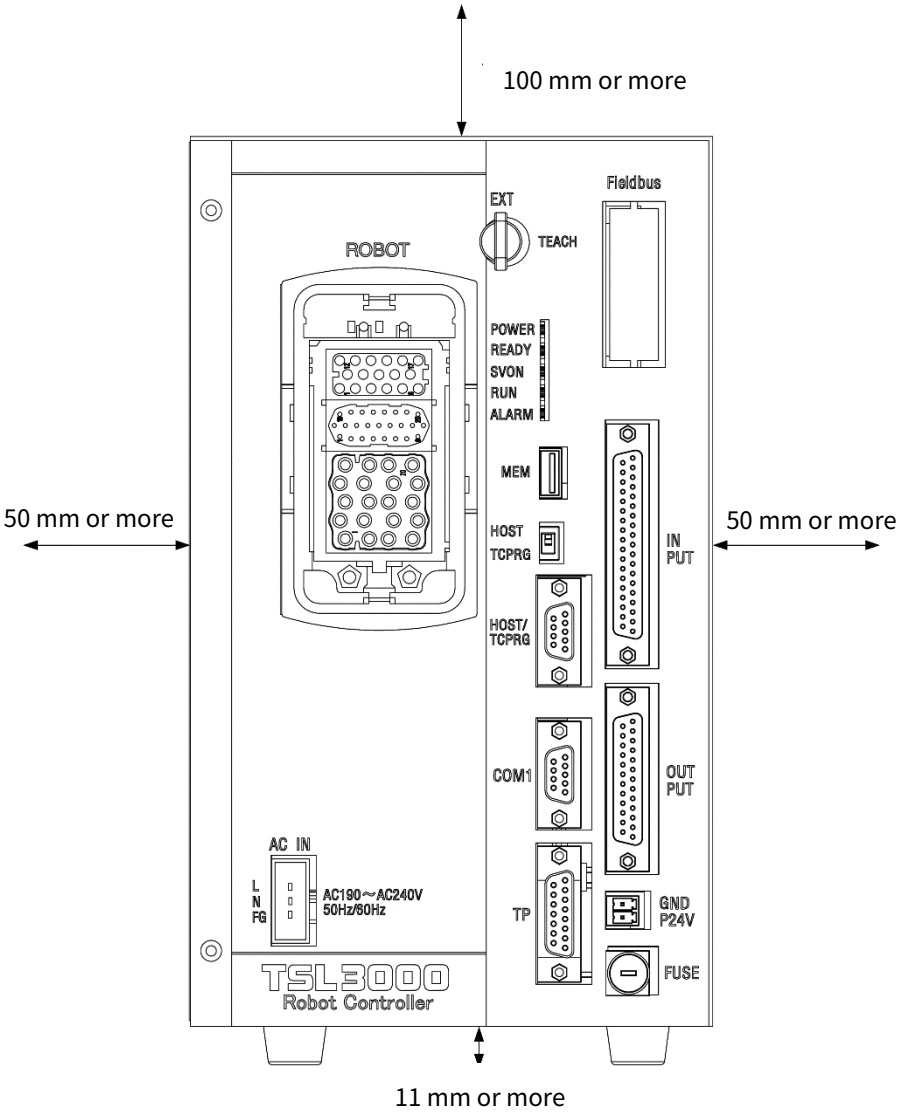



Figure 3.3 Air-passing spaces for the controller

## 3.4 Mounting Dimensions

When mounting the controller on the control panel, use the screw holes for rubber legs on the bottom to fit the mounting brackets and secure the controller to the control panel. The customer is requested to assemble them.

The mounting bracket (a pair of two) is optionally available as the unit code Y610A91A0. For consultation regarding the optional mounting bracket, contact our service or head office sales department.

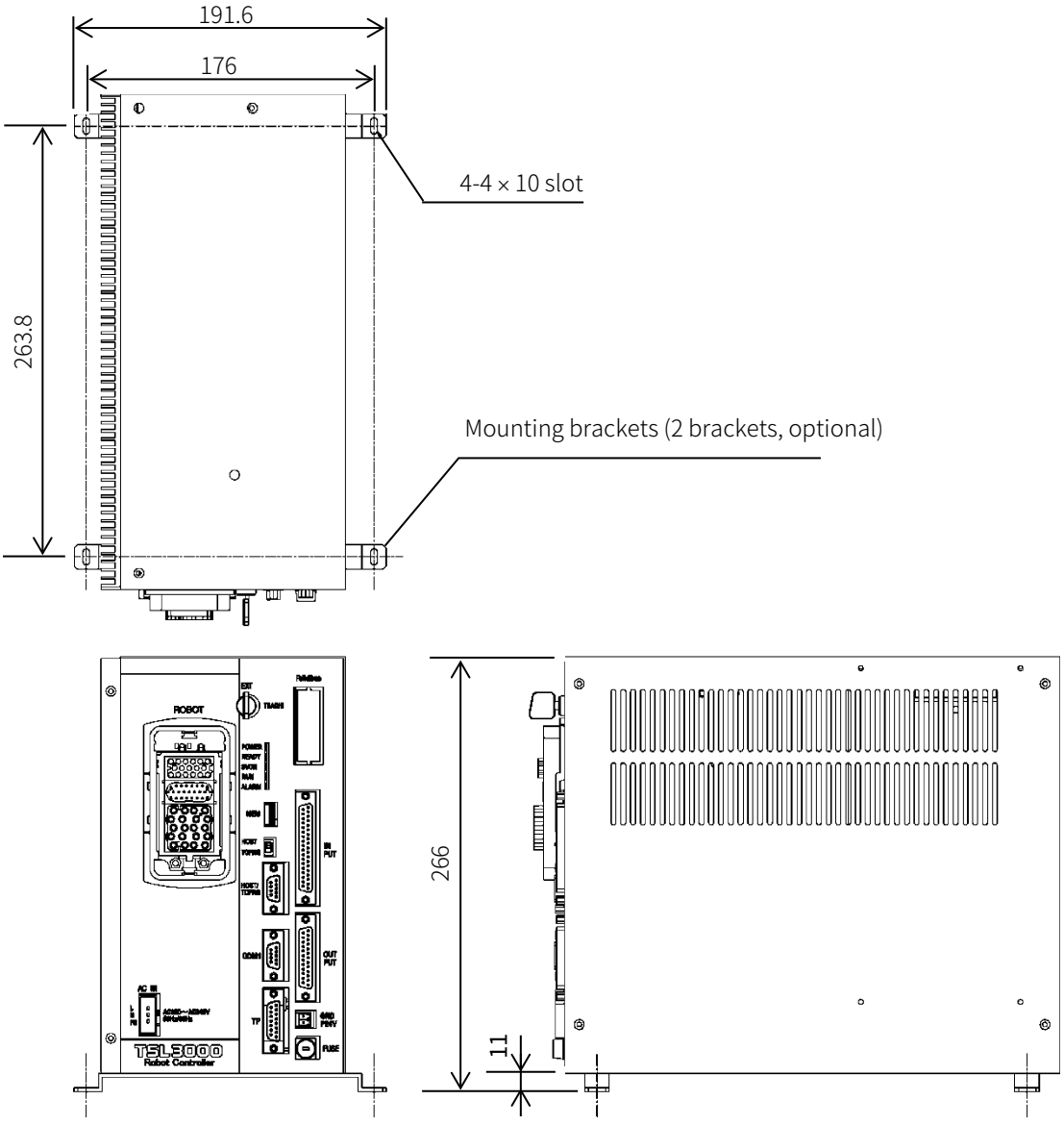



Figure 3.4 Screw hole dimensions for securing controller

## 3.5 Precautions for Mounting to the Control Panel

Observe the following when mounting the controller to the control panel, etc.

a) Remove the rubber feet from the bottom surface, and use these rubber foot mounting holes to mount the controller to the control panel.

b) Keep 110 mm clearance from the back of the controller for connecting cables for the Ethernet support and distributed I/O (option).

c) Remove the top cover when performing maintenance of the controller.

d) Consider serviceability for maintenance of the controller when installing the controller. When storing the controller to the control panel, ensure that the controller can be detached from the control panel for maintenance.

Specifically, consider the following.

1) Controller cable routing (Ensure that the controller can be pulled out.)

2) Ensure that the robot can be operated with all cables connected when the controller is pulled out of the control panel.

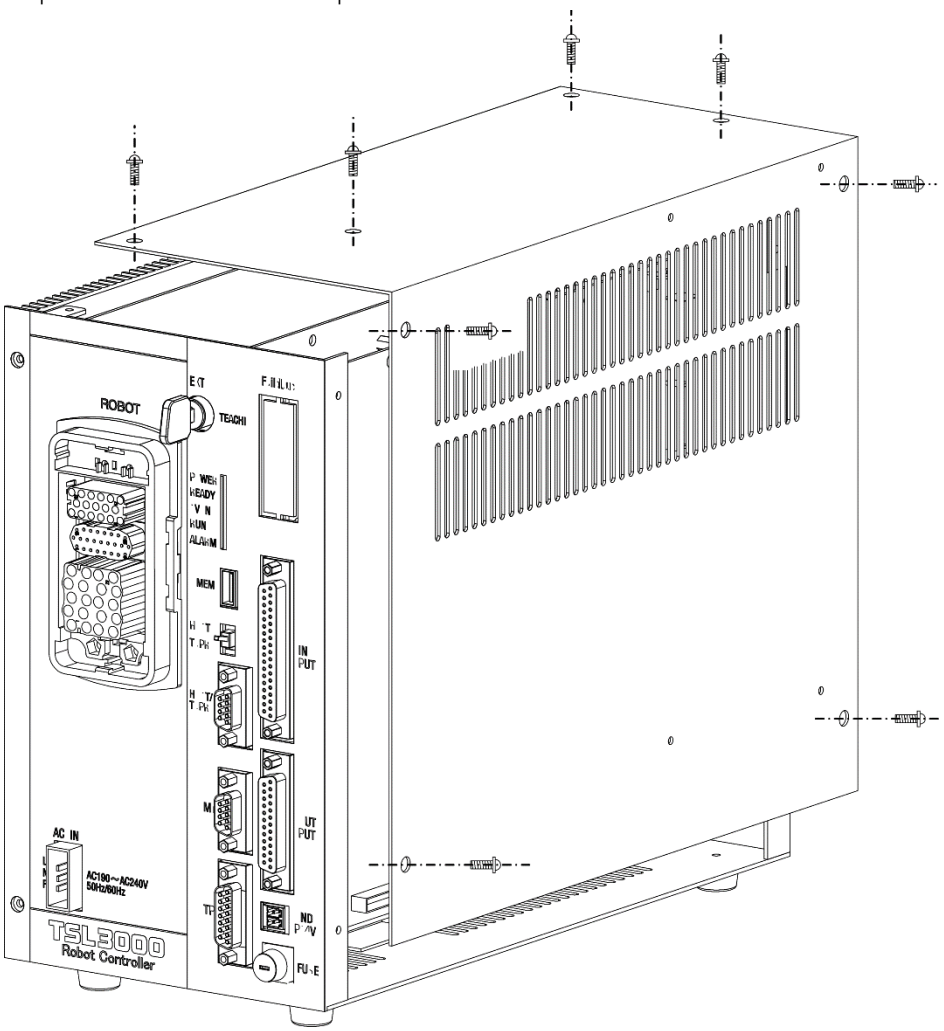



Figure 3.5 Removing the top cover

e) Keep clearance from the front of the controller for connecting robot cable connectors.

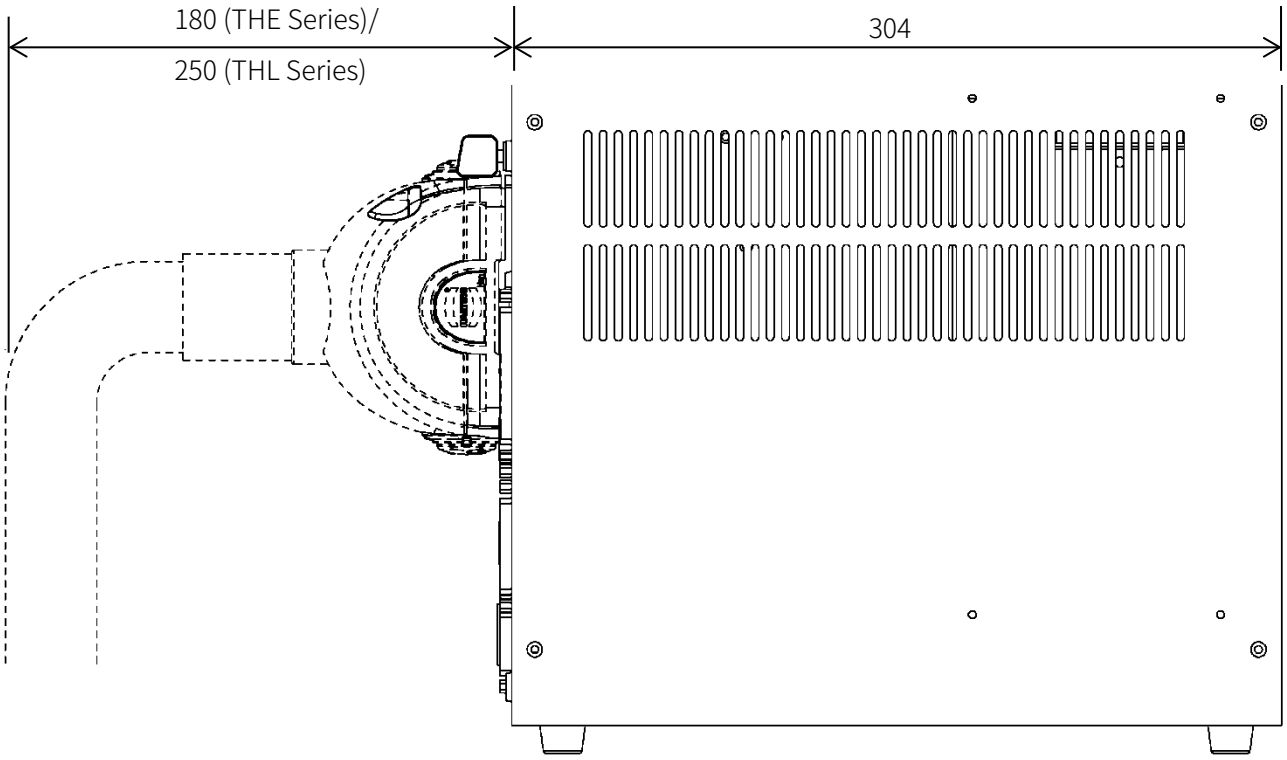



Figure 3.6 Clearance from the front of the controller

**NOTICE**

**Instruction**

- When the control panel is sealed completely, create vent holes to dissipate heat or install a fan to ventilate the controller, or apply other indirect cooling methods. Failure to do so will result in heat buildup in the control panel and controller, resulting in malfunction.

# 4 Cable Connection

Cable connection can only be carried out by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

**⚠DANGER**

| | |
|---|---|
|  <br>**Prohibited** | ● Never use wires other than those specified. Doing so will cause a fire.<br>● Do not make connections to connectors or terminals other than those specified. Doing so may cause a fire or an electric shock. |
|  <br>**Instruction** | ● Be sure to connect a ground wire. Failure to do so may cause electric leakage, resulting in electric shock or fire.<br>● After wiring, check the voltage and for short-circuit with a tester, or other means. If the wiring is incorrect or the controller is malfunctioning, turning on the power will cause a fire or an electric shock.<br>● Make sure that connectors are securely connected. If their connection is not made properly, static electricity may enter and cause malfunctions, which may cause you to be struck by the robot. |

| NOTICE | |
|---|---|
| Prohibited | ● Do not make connections to connectors or terminals other than those specified. Doing so may cause a malfunction. |

## 4.1 Wiring

This section describes types of cables and connectors and how to connect them.

### 4.1.1 Connectors on the controller

Figure 4.1 shows the cables connected to the controller.

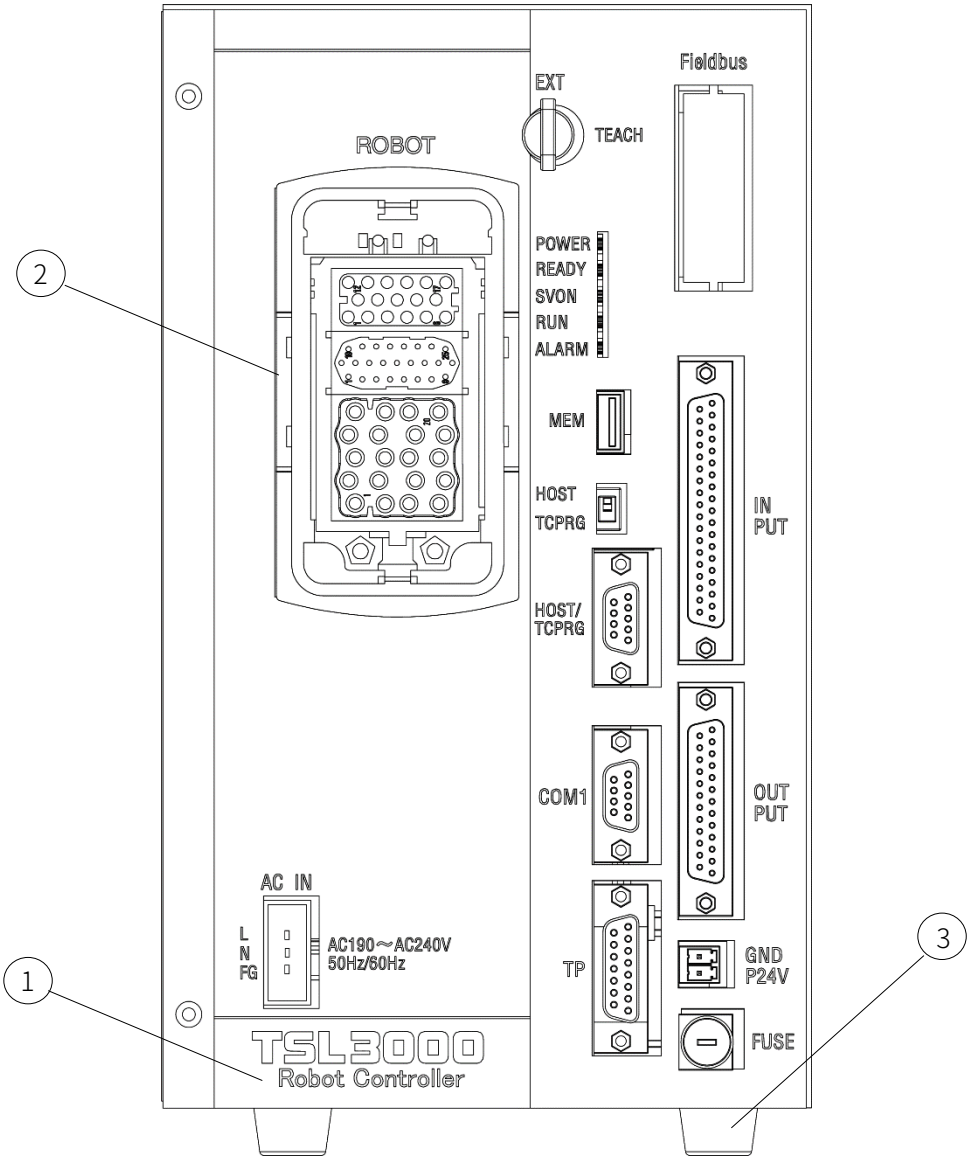



Figure 4.1 Controller connector layout

[1] "ACIN" connector for connecting the power cable
[2] "ROBOT" connector for connecting the motor, encoder, robot control signal, and brake signal cables
[3] "GNDP24V" connector for connecting the I/O signal power cable
The following sections describe connecting the above cables. For connecting other cables, refer to the ST85364 TSL3000 Instruction Manual for Interface.

### 4.1.2 "ACIN" connector for connecting the power cable (Figure 4.1-①)

The power cable is used to supply power to the controller. The ACIN plug connector (Figure 4.1 - ①) is supplied with the product. Cables are not supplied with the product. Customers need to provide a cable for the supplied connector to be connected to the controller side ACIN. For the terminal layout, see "4.1.9 Connector terminal layout list."

ACIN plug connector model:
03JFAT-SAYGF-I, Manufacturer: Japan Solderless Terminal MFG
Line: 0.8 $mm^2$ to 2.0 $mm^2$ (AWG#18 to AWG#14)

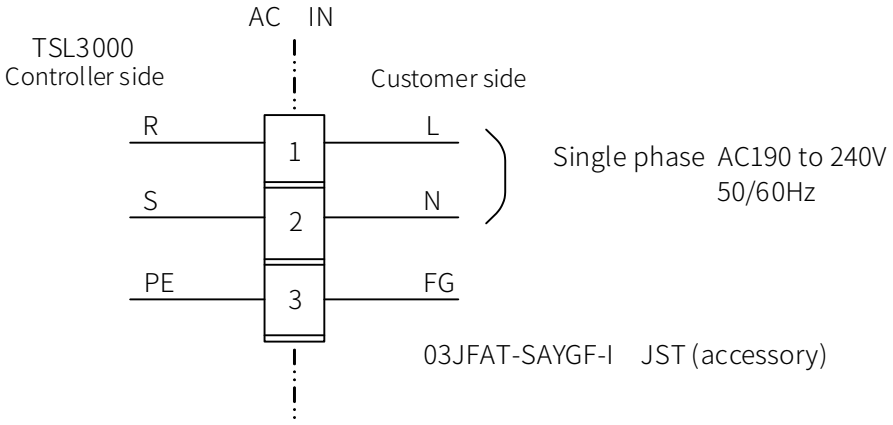



Table 4.1 Power supply specifications

| | |
|---|---|
| Power supply | Single-phase, 190 to 240 VAC<br>50/60 Hz ± 1 Hz |
| Momentary power loss | Within 40 msec |
| Grounding | Class D only |

**⚠ DANGER**

**Prohibited**

- Never insert conductive objects or your fingers into the power supply terminal. Doing so may cause an electric shock.

> ***MEMO***
>
> *"8-027 Slow charge error" occurs at servo ON if power is not supplied properly to the controller due to an open phase or voltage drop. In this case, confirm that the power supply voltage at the AC power supply connector is stable and satisfies the controller input power supply specification value. For details of "8-027 slow charge error", see the error codes in "ST80723 TS3000 for Alarms".*

### 4.1.3 "ROBOT" connector for connecting the motor cable (Figure 4.1-②)

The motor cable, together with the encoder cable, robot control signal cable, and brake signal cable, are integrated into a single bundle that is coupled with the robot. The motor cable connected between the controller and the robot supplies power for motors to drive from the controller servo driver to each axis motor.

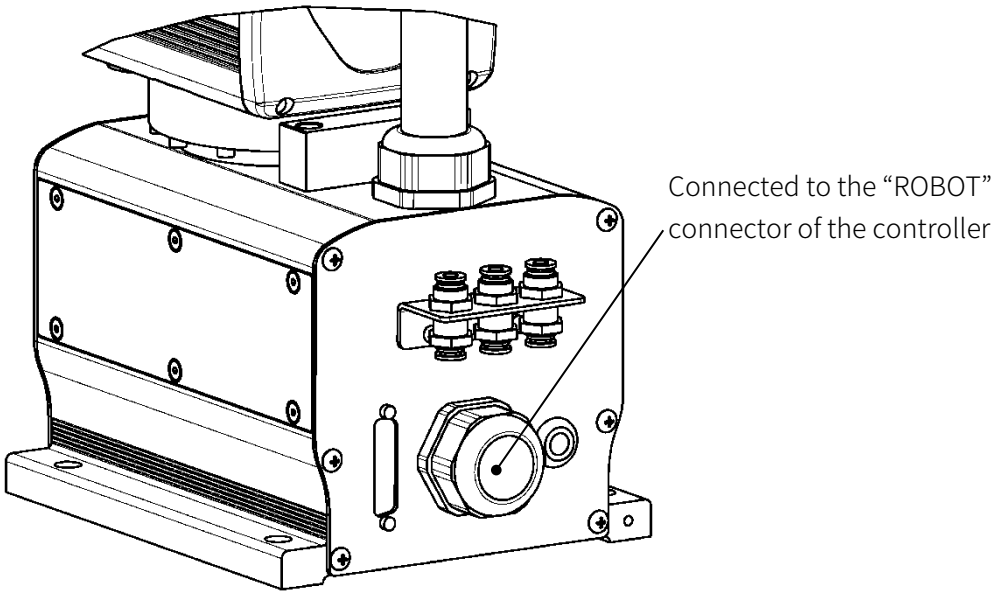



Figure 4.2 Robot-side connector layout (THL Series)

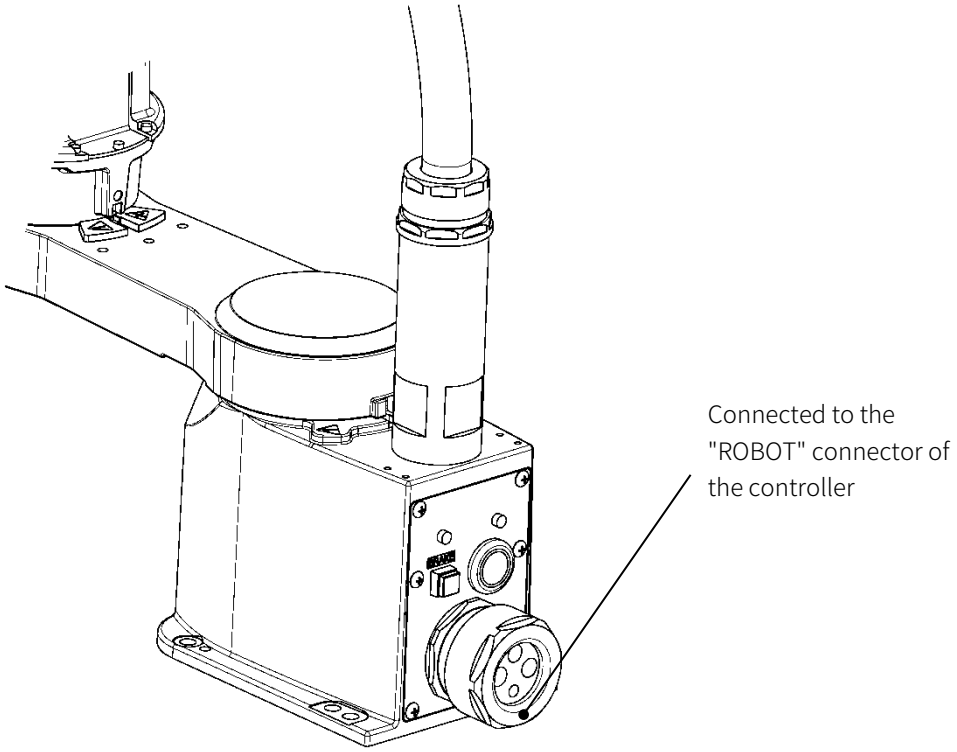



Figure 4.3 Robot-side connector layout (THE Series)

### 4.1.4 "ROBOT" connector for connecting the encoder cable (Figure 4.1-②)

The encoder cable, together with the motor cable, robot control signal cable, and brake signal cable, are integrated into a single bundle that is coupled with the robot. The encoder cable is a signal line to input signals from the rotation angle detection encoder of each axis of the robot to the controller.

### 4.1.5 "ROBOT" connector for connecting the robot control signal cable (Figure 4.1-②)

The robot control signal cable, together with the motor cable, encoder cable, and brake signal cable, are integrated into a single bundle that is coupled with the robot. The robot control signal cable is used to input/output robot control signals such as the motor axis fixing brake ON/OFF signal and hand operation signal.

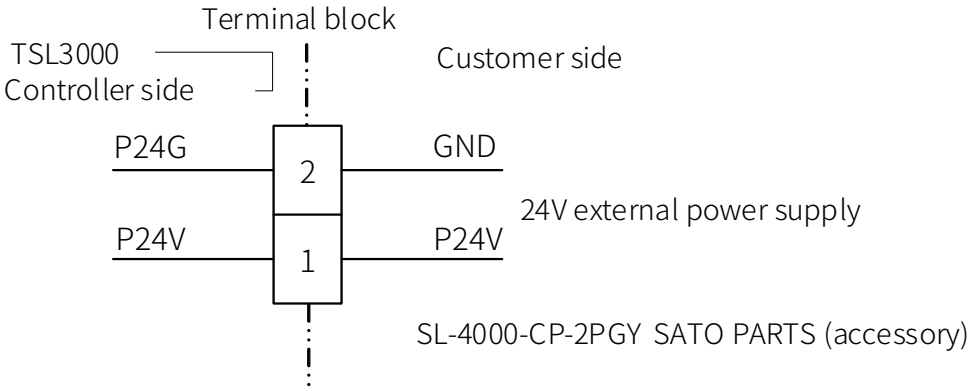
### 4.1.6 "ROBOT" connector for connecting the brake signal cable (Figure 4.1-②)

The brake signal cable, together with the motor cable, encoder cable, and robot control signal cable, are integrated into a single bundle that is coupled with the robot. The brake signal cable is used to turn on/off the motor axis fixing brake.

### 4.1.7 "GNDP24V" connector for connecting the I/O signal power cable (Figure 4.1-③)

Use the accompanying connector SL-4000-CP-2PGY to connect the I/O signal power cable. The P24V power supply is provided from the front of the controller.
The following inputs/outputs use the external power supply (24 VDC). Ensure that the power is supplied from an external power supply (24 VDC).

- External I/O
- External operation I/O
- Extended I/O
- Hand I/O




Use cables with the size "AWG24 to 16" for connectors.
Select an appropriate external power supply in accordance with your system specifications such as the current capacity.
For connecting/disconnecting the I/O signal power cable, refer to the ST85364 TSL3000 Instruction Manual for Interface.

**NOTICE**

| | |
|---|---|
| **Instruction** | ● Apply a voltage within the range of 24V±10% to the I/O signal power cable. Applying a voltage exceeding the allowable value will damage the controller. |

### 4.1.8 Connecting/disconnecting the connectors

| ⚠WARNING | |
|---|---|
| **Instruction** | ● Turn off the power supply before disconnecting/connecting the cable from/to the controller. Touching the terminals of the disconnected cable while the power is on will cause electric shock.<br>● When disconnecting the cable, grasp the connector and pull it out while holding the controller. Pulling the cable will break the wire and cause an electric shock. |

| NOTICE | |
|---|---|
| **Instruction** | ● Turn off the power supply before disconnecting/connecting the cable from/to the controller. Disconnecting/connecting the cable while power is on will damage ICs on the board.<br>● When disconnecting the cable, grasp the connector and pull it out while holding the controller. Failure to do so will cause the controller to fall over, resulting in a malfunction. |

• ROBOT connector "ROBOT"

To connect the ROBOT connector, securely insert the cable connector into the connector on the controller side while pressing the top and bottom buttons on the cable connector. Loose insertion of the connector causes an accident due to poor contact of the connector; therefore, be sure to check whether the connector is securely connected.

To disconnect the ROBOT connector, follow the same procedure in reverse order. Pull out the cable connector while pressing the top and bottom buttons.

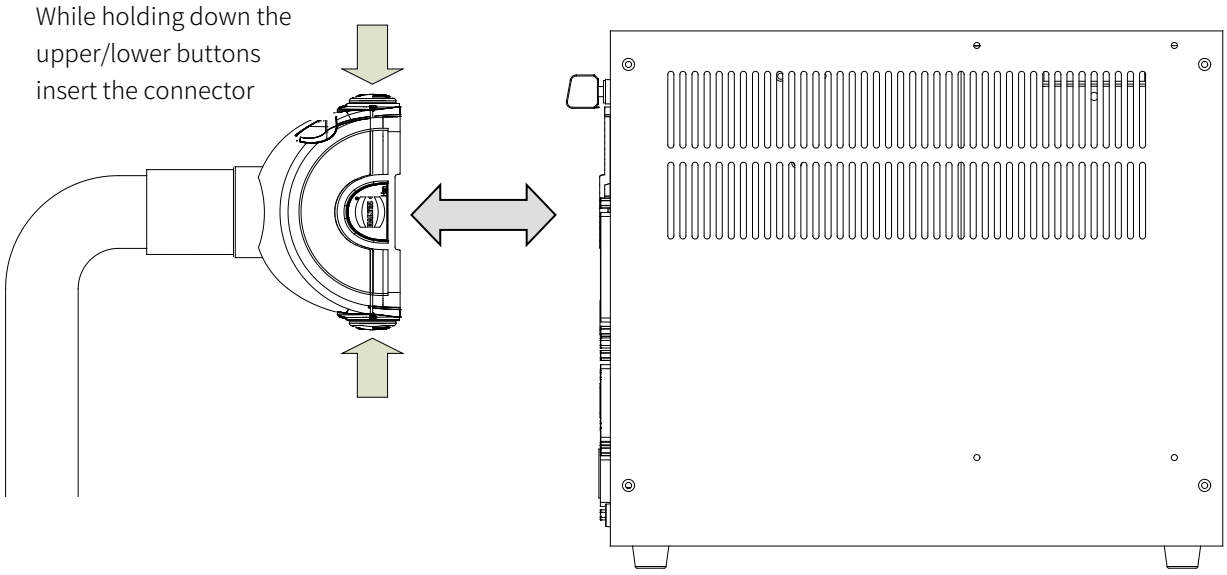
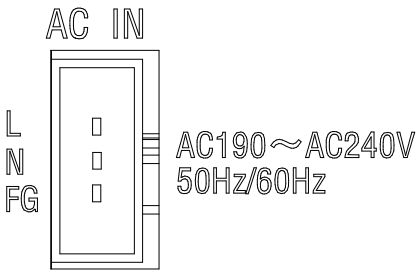



Figure 4.4 Connecting/disconnecting the ROBOT connector

### 4.1.9 Connector terminal arrangement

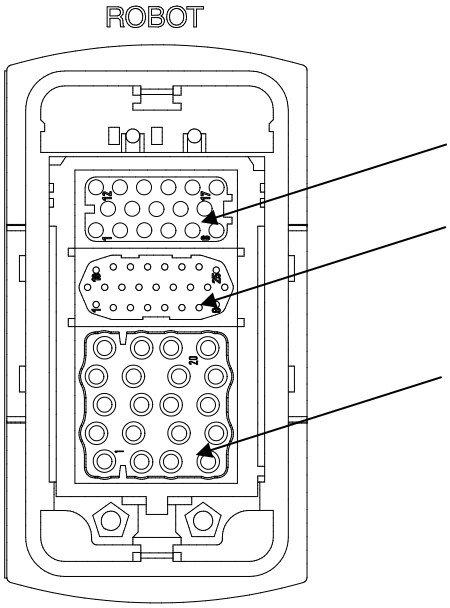
a) "ACIN" connector for power supply (Figure 4.1 - ①).




Controller side
Model: 03JFAT-SAYGF-I
Manufacturer: J.S.T. Mfg. Co., Ltd.
The paired power supply connector is also supplied.

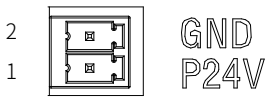
1(L)
2(N) — Single-phase 190 to 240 VAC, 50/60 Hz
3(FG) — Grounding (Class D or higher)

b) "ROBOT" connector for the motor, encoder, robot control signal, and brake signal cables (Figure 4.1 - ②)




Controller side
Manufacturer: HARTING

c) Connector for the power supply for external I/O signals "GNDP24V" (Figure 4.1-③)




Controller side
Model: SL-4000-CWJH 02PGY
Manufacturer: SATO PARTS

SATO PARTS
Parts for the device to be connected are supplied with the controller.
For details, refer to ST85364 TSL3000

## 4.2 Controller Connector Signals

### 4.2.1 Connector signal connection diagram

For detailed view of each connector terminals, refer to the ST85364 TSL3000 Instruction Manual for Interface, Chapter 2 "Connecting the External Cables".

### 4.2.2 Signal jumper connector for safety measures

The following system input signal lines are provided as safety measures.
The robot will not operate unless jumper connections for the following signal lines are made.
For this purpose, this controller is supplied with an INPUT jumper connector (part name: I/O jumper connector) with the following system input signals jumper-connected as standard. When this connector is used, the internal jumper is set to P24V (+) common Type N.

| | | |
|---|---|---|
| System input signal lines ... | INPUT-12 | STOP) |
| | INPUT-14 | (SVOFF) |
| | INPUT-32 | (BREAK) |
| | INPUT-18, 19 | (EMS1B to EMS1C) |
| | INPUT-36, 37 | (EMS2B to EMS2C) |
| | INPUT-17 | (INCOM) |
| | INPUT-35 | (P24V for type N (X8YXA), P24G for type P (X8YXB)) |

Jumpers of supplied connector

| INPUT | | | | | |
|---|---|---|---|---|---|
| 12-16 | 14-16 | 32-16 | 17-35 | 18-19 | 36-37 |

Also, apply jumpers to the following signals if they are not used as system signals.

INPUT-13 (LOW_SPD)
INPUT-31 (CYCLE)

***MEMO***

1. *If the "SVOFF" signal and emergency stop contacts 1 and 2 are not jumped, the servo power supply of the controller cannot be tuned on.*
2. *If the "CYCLE" is not jumped, the controller will switch to the cycle operation mode.*
3. *If the "LOW_SPD" signal is not jumped, the robot automatic operation speed will be slow.*
4. *If the "STOP" signal is not jumped, the robot automatic operation will not be available.*
5. *If the "BREAK" signal is not jumped, the robot automatic operation will not be available.*

# Maintenance

This part describes the maintenance and inspection of the controller.

# 5 Maintenance

The maintenance and inspection of the controller can be carried out only by qualified personnel.
Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

Recommended protector:

| Type and name | Protection part and use | Recommended example |
|---|---|---|
| Helmet | Protection part: Head<br>Use: Protect from a falling object.<br>Protect from collision with the arm. |  |
| Safety glasses | Protection part: Eyes<br>Use: Protect from a flying object.<br>Protect from collision with the arm. |  |
| Protective gloves | Protection part: Hands and fingers<br>Use: Protect hands and fingers when caught in the machine.<br>Prevent a carried object from dropping. |  |
| Protective shoes | Protection part: Feet and toes<br>Use: Protect from a falling object. |  |

**⚠WARNING**

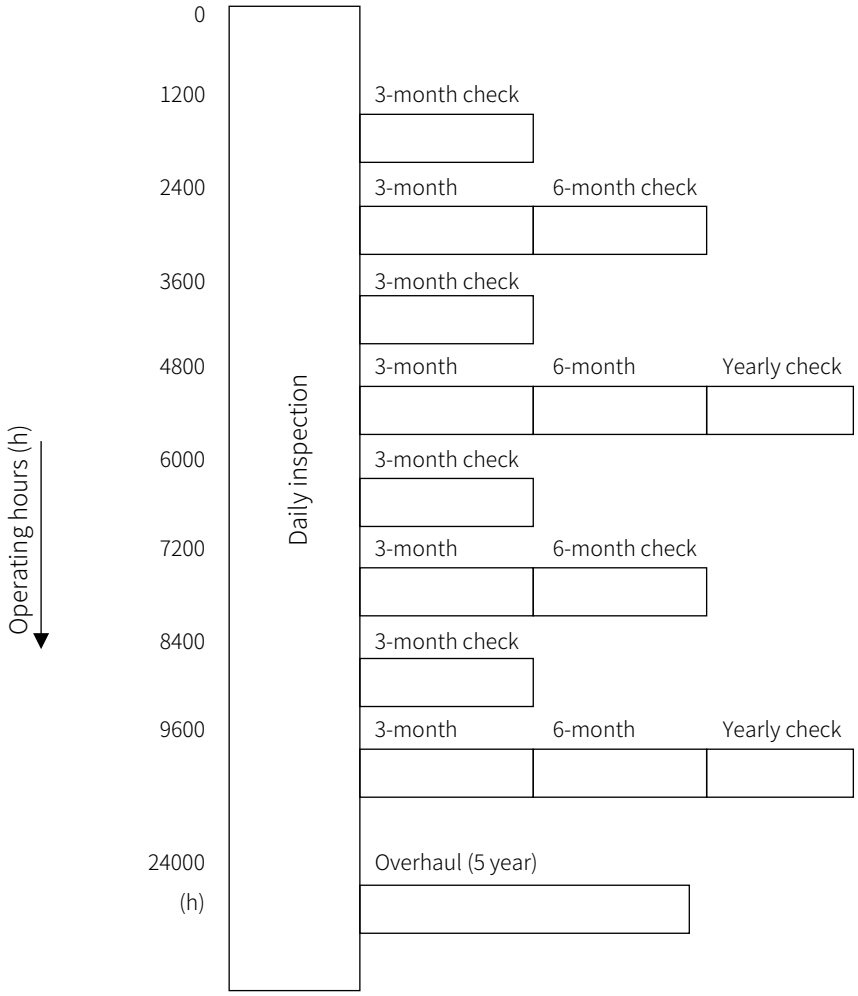


**Instruction**

- Conduct maintenance and inspection on a regular basis. Failure to do so may cause accidents.

## 5.1 Maintenance Schedule

Daily and periodic maintenance and inspection are required. Inspection items are added in each periodic maintenance and inspection carried out every 1200 operating hours.




Periodic maintenance and inspection time calculation

16 operating hours per day: 1200 hours / 16 hours / 25 days = 3 months
24 operating hours per day: 1200 hours / 24 hours / 25 days = 2 months

If the operating hours per day are 16, maintenance and inspection are required every three months. If the operating hours per day are longer than 16, earlier maintenance and inspection are required.

Figure 5.1 Maintenance schedule

## 5.2 Maintenance/Inspection Items

Daily and periodic maintenance and inspection are required.
This section describes items of maintenance and inspection. For details of each maintenance and inspection, refer to the relevant section stated in the reference column.

### 5.2.1 Inspection in the power-off state

Table 5.1 Inspection in the power-off state

| Inspection | Location | Daily check | 3-month check | 6-month check | Yearly check | Reference |
|---|---|---|---|---|---|---|
| Check screws for tightness. If they are loose, retighten them. | Cover mounting screws | ○ | ○ | ○ | ○ | 6.3.4<br>6.3.5<br>6.3.6 |
| Connector tightness | All connector section | ○ | ○ | ○ | ○ | 4.1.8 |
| Damage check<br>Cleaning of accumulated dust | The entire robot controller | ○ | ○ | ○ | ○ | |
| Vent hole check | Vent holes | ○ | ○ | ○ | ○ | 6.3.1 |

### 5.2.2 Inspection in the power-on state

Table 5.2 Inspection in the power-on state

| Inspection | Location | Daily check | 3-month check | 6-month check | Yearly check | Reference |
|---|---|---|---|---|---|---|
| Emergency stop validation | Emergency Stop button | ○ | ○ | ○ | ○ | 6.3.2<br>6.3.3 |
| Enable switch validation | Enable switch | ○ | ○ | ○ | ○ | |
| Mode switch<br>TEACH/EXT mode validation | TEACH/EXT mode | ○ | ○ | ○ | ○ | *1 |
| Robot/controller validation | All axis validation in TEACH mode | ○ | ○ | ○ | ○ | *2 |

*1: Refer to ST80720 TS3000 Instruction Manual for Operation, Section 1.2.1.
*2: Refer to ST80720 TS3000 Instruction Manual for Operation, Section 3.1.

### 5.2.3 Overhaul

Table 5.3 Overhaul

| Maintenance details | Service parts | 5 years | Reference |
|---|---|---|---|
| Battery replacement | Battery for controller backup | ○ | 6.3.4 |
| Switching power supply replacement | Inside the controller | ○ | 6.3.5 |

## 5.3 Maintenance Contract and Repair Service

### 5.3.1 Maintenance contract

Inspections after the six-month maintenance are difficult for customers to perform. Please request us to provide such inspections. We recommend that you make a service agreement with us at the time of purchase.

### 5.3.2 Repair service

If the system malfunctions, turn off the power supply, and contact our service department. Please notify us of the details of the failure and the following information stated on the nameplates of the robot and controller. (For the contact information of our service department, refer to "Service Network".)

- Robot model
- Robot product number
- Controller product number
- Robot production date

For the product number and production date, check the nameplate on the robot base.

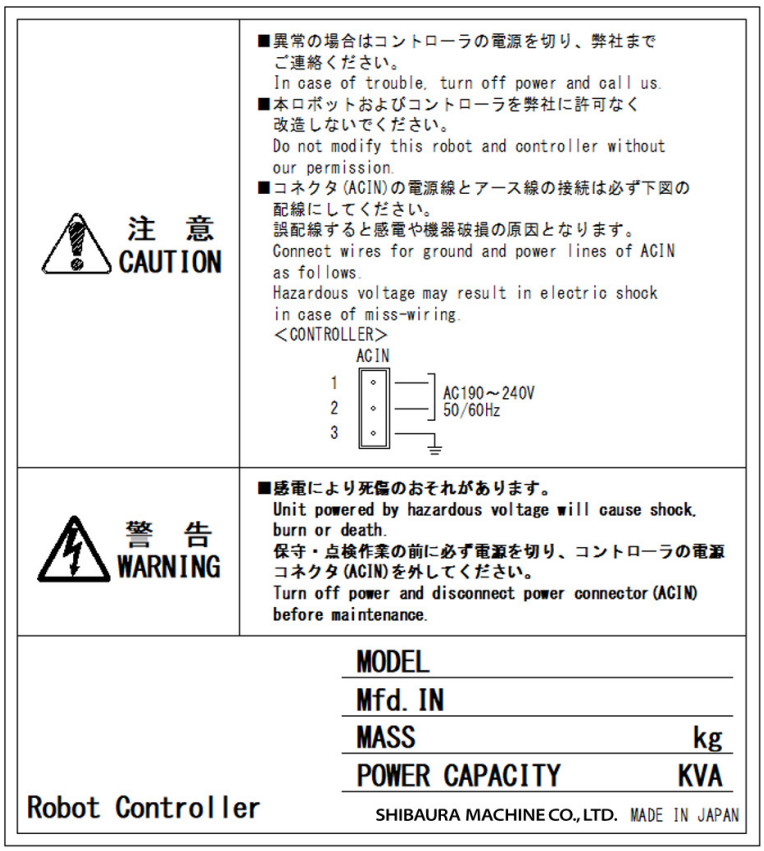



Figure 5.2 Nameplates

### 5.3.3 Modification

Do not disassemble or modify the controller without consent of SHIBAURA MACHINE.

| ⚠ DANGER | |
|---|---|
|  **Prohibited** | ● Never replace or modify parts that are not stated in the instruction manual.<br>• Doing so will cause performance degradation or malfunction resulting in electric shock.<br>• Doing so will cause performance degradation or malfunction resulting in a fire.<br>• Doing so will cause you to be struck by the robot moving unexpectedly. |

## 5.4 Maintenance Tools

We recommend that you provide the following tools for maintenance.

- Phillips screwdrivers #1, #2, flat head screwdriver #2
- Nippers (to cut cable ties)

Recommended items other than tools

- Alcohol (for cleaning), cloth
- TSAssist program support software (option)

# 6 Maintenance of the Controller

The maintenance and inspection of the controller can be carried out only by qualified personnel. Also, be sure to follow the laws and regulations as well as ordinances of the applicable country.

## 6.1 Precautions for Maintenance and Inspection

For maintenance and inspection of the controller, observe the following to ensure safety.

| ⚠ DANGER | |
|---|---|
|  **Prohibited** | ● When the controller cover is opened, do not allow water or other liquids or foreign objects to enter the inside of the controller. Doing so may cause a short circuit inside the controller and result in a fire or an electric shock. |
|  **Instruction** | ● For some maintenance and replacement parts, our dedicated parts must be used. The use of connectors and terminals other than those specified will result in a fire or an electric shock.<br>● For maintenance and replacement not described in this manual, contact our service company. The maintenance/replacement service is available for a fee. We shall not provide any warranty for accidents or damages caused by the replacement work performed by the customer or as a result thereof. |

| ⚠WARNING | |
|---|---|
|  **Prohibited** | ● Do not touch the controller terminals or internal circuit board for 2 minutes after removing power. Failure to do so will result in electric shock from the voltage remaining in the large capacitor built into the controller. |
|  **Instruction** | ● When replacing parts, use spare parts specified by us. Use of parts other than those specified can result in a fire or cause you to be struck by the robot moving unexpectedly.<br>● Before starting work, be sure to wear protective equipment and unplug the power supply. If you start work with the power on, you will receive an electric shock from the power board, servo board, switching power supply, etc. that are electrified.<br>● Before opening the cover of the controller, turn off the power. Opening the cover while the power is on will cause an electric shock.<br>● Before entering the movable range of the robot, follow the instructions below.<br>• Wear protective equipment.<br>• Keep the TP within the reach of the operator.<br>• Change the mode to TEACHING mode, and pull out the mode switching key so that the mode cannot be changed.<br>⇒ Failure to follow the instructions above will allow the emergency stop to be canceled by a third party and cause you to be struck by the robot during routine inspection work within the moving range of the robot.<br>● Check if PN voltage remains in the capacitor by measuring the voltage and checking the LED.<br>If PN voltage remains in the capacitor, you will receive an electric shock.<br>● Do not use metal scales while the controller is energized. Doing so and contacting the current-carrying parts may cause electric shock. |

| NOTICE | |
|---|---|
|  **Instruction** | ● For some maintenance and replacement parts, our dedicated parts must be used. The use of connectors and terminals other than those specified will result in failure. |

## 6.2 Controller Component Layout

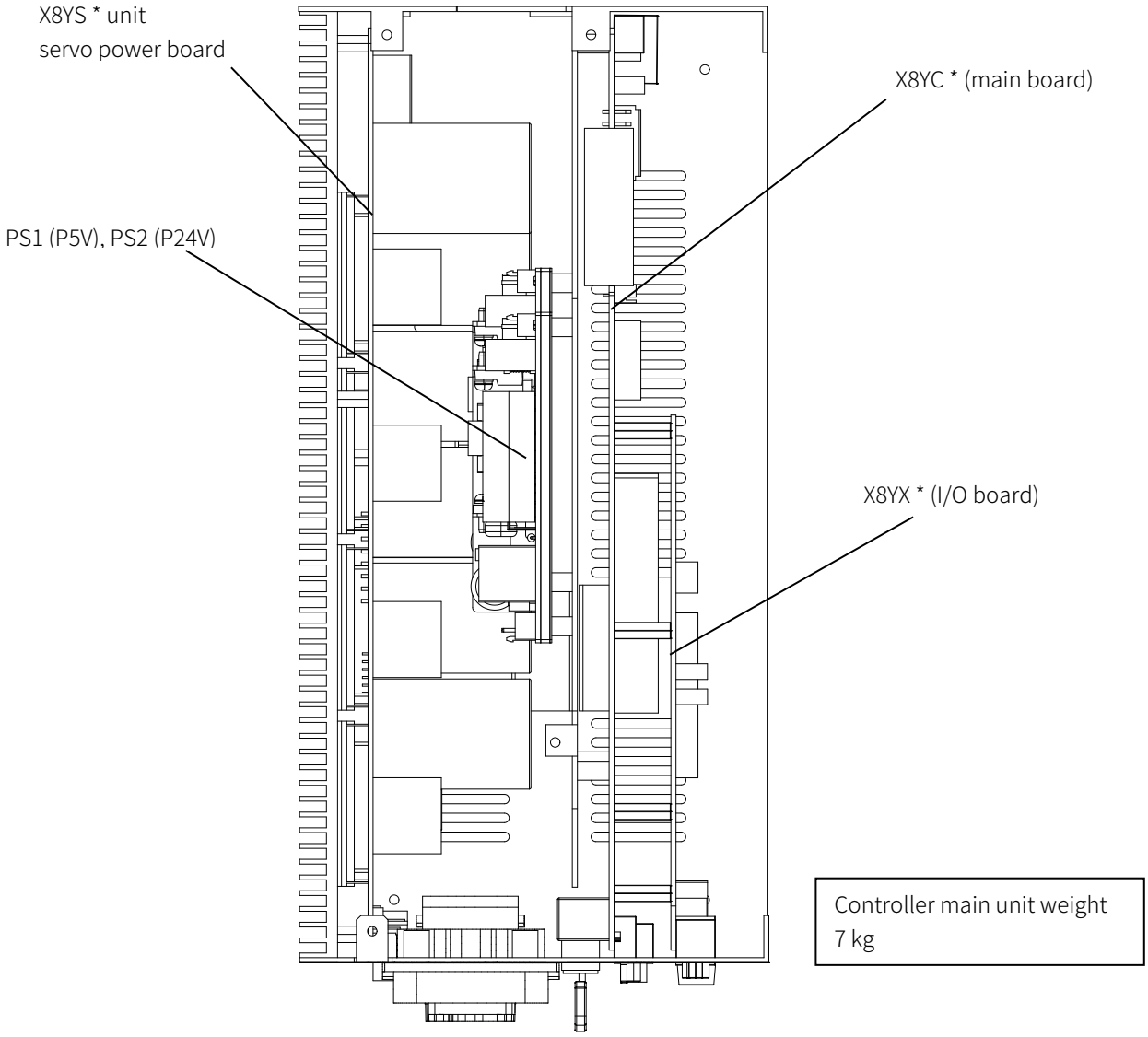



Figure 6.1 Controller component layout

Table 6.1 Controller components

| Name | Description |
|---|---|
| PS1, PS2 | PS1(P5V), PS2(P24V) output Switching power supply |
| X8YC* | Main control board |
| X8YX* | I/O Output board |
| X8YS* | Servo power supply module |

## 6.3 Maintenance Procedure

### 6.3.1 Checking the vent holes of the controller

If the vent holes are blocked, the temperature inside the controller increases, resulting in malfunction. Check the following to ensure that the vent holes are not blocked.

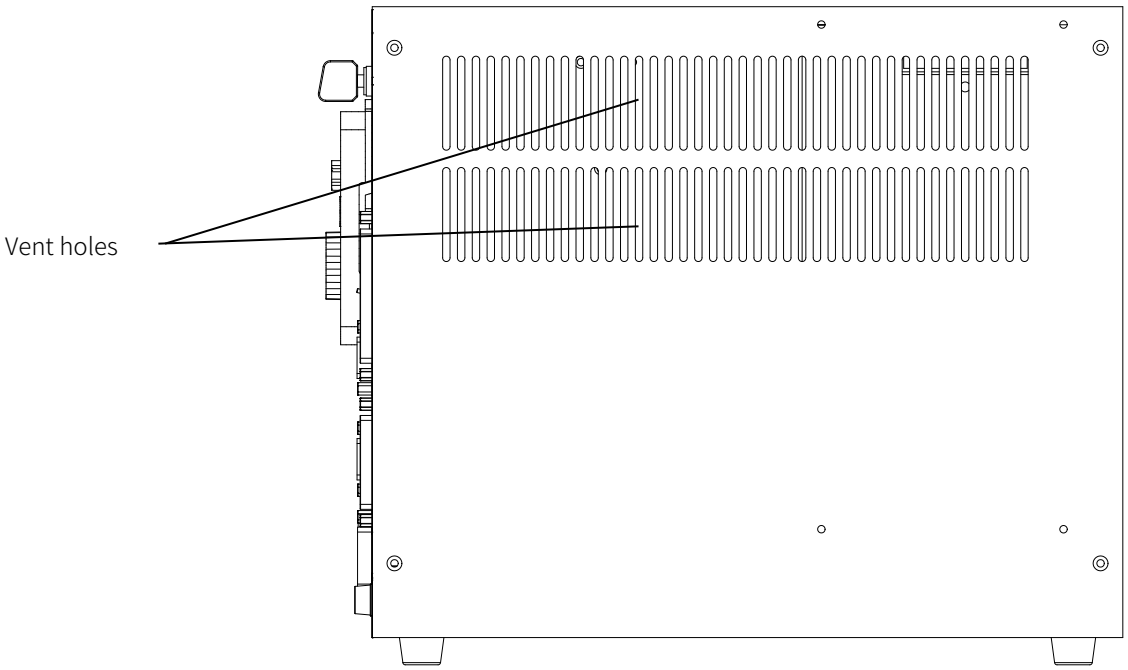



Figure 6.2 Rear view of the controller

1) Remove any object in front of the vent holes to ensure ventilation.
2) Ensure there is no substance stuck in the vent holes.
   Remove any substance stuck in the vent holes.

### 6.3.2 Safety device validation (TP1000, TP1000-6AX)

Confirm that the emergency stop switch and other switches of the teaching pendant and safety devices connected to the external operation input signals function normally.

1) Confirm that the emergency stop switch of the teaching pendant functions normally.
   a) Turn the key on the front of the controller to the TEACH position.
   b) Turn the emergency stop switch ① clockwise and confirm that it turns off.
   c) While holding the Enable switch ③ of the teaching pendant at its neutral position, press the Servo ON switch ② to turn on the servo. Confirm that the LED of the Servo ON switch ② turns on.
   d) Press the emergency stop switch ① of the teaching pendant and confirm that the servo is turned off. The LED of the Servo ON switch ② turns off.
      * In this case, confirm that the emergency stop switch ① is in the pressed state.
   e) Turn the emergency stop switch ① clockwise and confirm that it turns off.

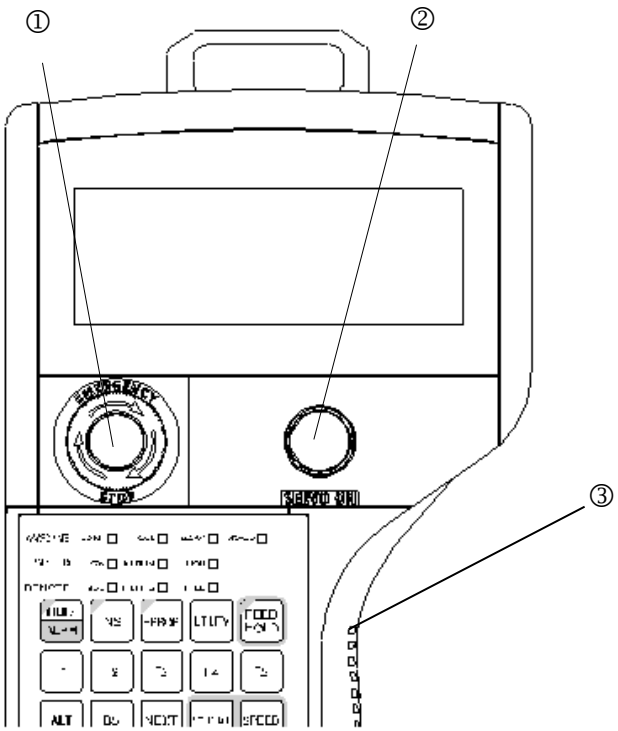



Figure 6.3 Emergency stop switch of the teaching pendant (TP1000)

2) Confirm that safety devices connected to the external operation input signals function normally.
   a) Connect the power plug to the power supply, turn on the servo, turn on the safety devices (external emergency stop switch, beam type safety device, foot switch, etc.) connected to the "emergency stop" external operation input signals, and confirm that the servo can turn off.
   b) Confirm that safety devices connected to the "low speed instruction" external operation input signals turn on and function normally.

### 6.3.3 Safety device validation (TP3000)

Confirm that the emergency stop switch and other switches of the teaching pendant and safety devices connected to the external operation input signals function normally.

1) Confirm that the emergency stop switch of the teaching pendant functions normally.
   a) Turn the key on the front of the controller to the TEACH position.
   b) Turn the emergency stop switch ① clockwise and confirm that it turns off.
   c) While holding the Enable switch ③ of the teaching pendant at its neutral position, press the Servo ON switch ② to turn on the servo. Confirm that the LED of the Servo ON switch ② turns on.
   d) Press the emergency stop switch ① of the teaching pendant and confirm that the servo is turned off.
      The LED of the Servo ON switch ② turns off.
      * In this case, confirm that the emergency stop switch ① is in the pressed state.
   e) Turn the emergency stop switch ① clockwise and confirm that it turns off.

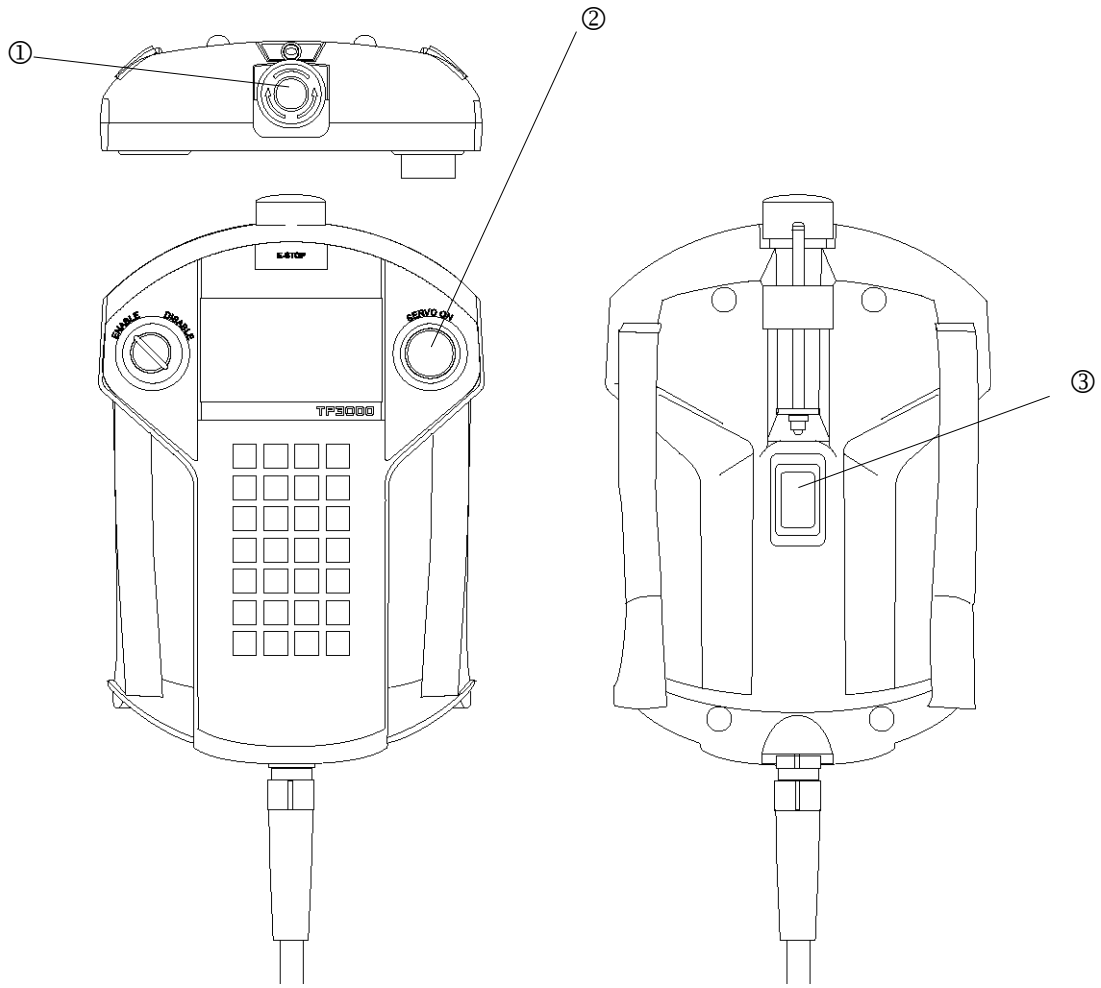



Figure 6.4 Emergency stop switch of the teaching pendant (TP3000)

2) Confirm that safety devices connected to the external operation input signals function normally.
   a) Connect the power plug to the power supply, turn on the servo, turn on the safety devices (external emergency stop switch, beam type safety device, foot switch, etc.) connected to the "emergency stop" external operation input signals, and confirm that the servo can turn off.
   b) Confirm that safety devices connected to the "low speed instruction" external operation input signals function normally.

### 6.3.4 Battery replacement

**DANGER**



**Prohibited**

- Never incinerate, disassemble, short circuit, or charge batteries. Doing so will cause rupture, resulting in a fire. Dispose of batteries according to your company's rules.

**NOTICE**



**Instruction**

- The battery replacement work should be completed within 20 minutes. Failure to do so will result in the loss of all memory contents.

The memory installed on the X8YC board of the controller is backed up by the lithium battery to retain data in the memory.
Replace the battery every five years. The lithium battery has a certain lifespan. Using beyond the lifespan causes battery voltage drop below the level required for retaining data in the memory. In this case, data will be lost and malfunction may result due to liquid leakage.
This replacement work is difficult to be performed by the customer. Contact our service department.

Replace the battery if any of the following alarms occurs.
"MAIN Battery alarm" (Alarm code 1-145)

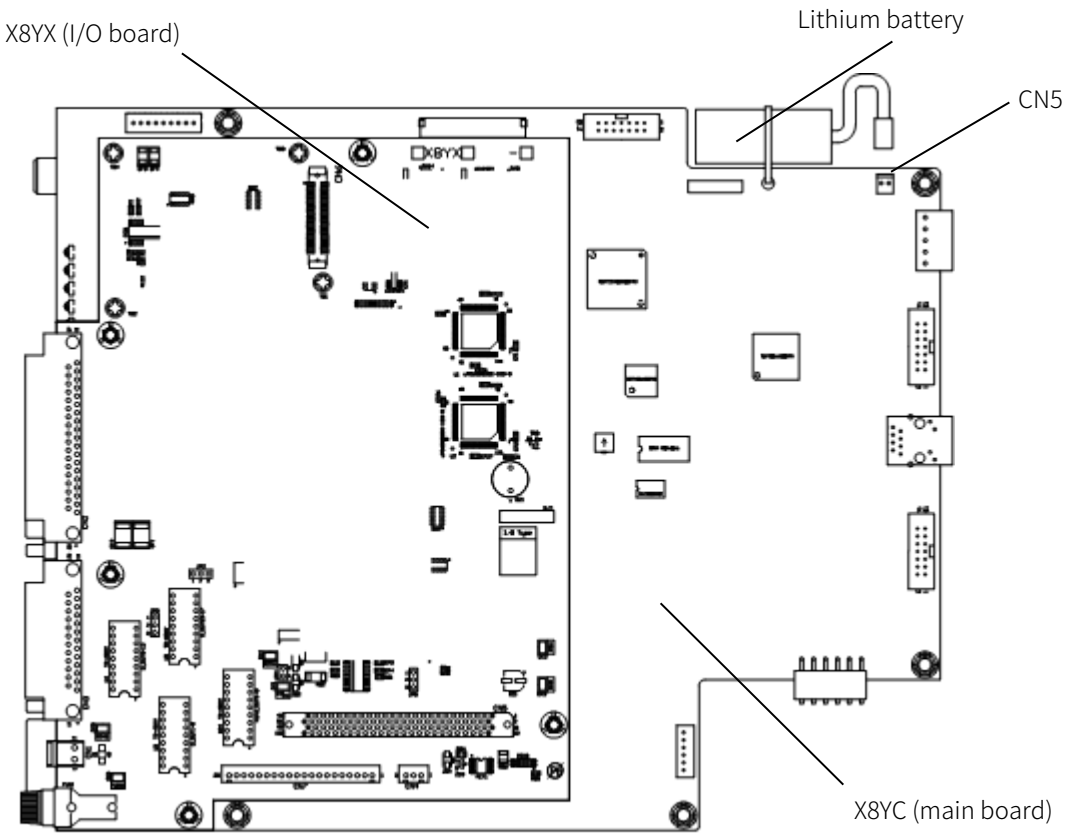



Figure 6.5 Lithium battery location

Table 6.2 Battery replacement

| Battery type | Manufacturer | Battery life | Remarks |
|---|---|---|---|
| ER6C WK27 | HITACHI MAXELL | 10 years (normal room temperature) | Discontinued in 2022 |
| ER6V C4 | TOSHIBA | 10 years (normal room temperature) | Adopted from 2022 |

Note: The battery life is 10 years at normal temperature, but it varies depending on the operating temperature, humidity, and other environmental factors. We recommend replacing the battery every five years.
The dedicated battery specified in the above table must be used. Contact us to purchase the battery.

Replacement procedure

1) If the controller is to be left without the battery for twenty minutes or more, save the programs and parameters from the memory to the computer in advance. Data in the memory will be lost.
2) Turn off the breaker for the power supply, and remove the power plug from the power supply.
3) Remove the ACIN connector from the controller.
4) Prepare a new battery, remove the eight flat head screws (M3 × 6) fixing the cover, and remove the cover from the unit.
   (* Mount the cover firmly by tightening the screws after the battery is replaced.)

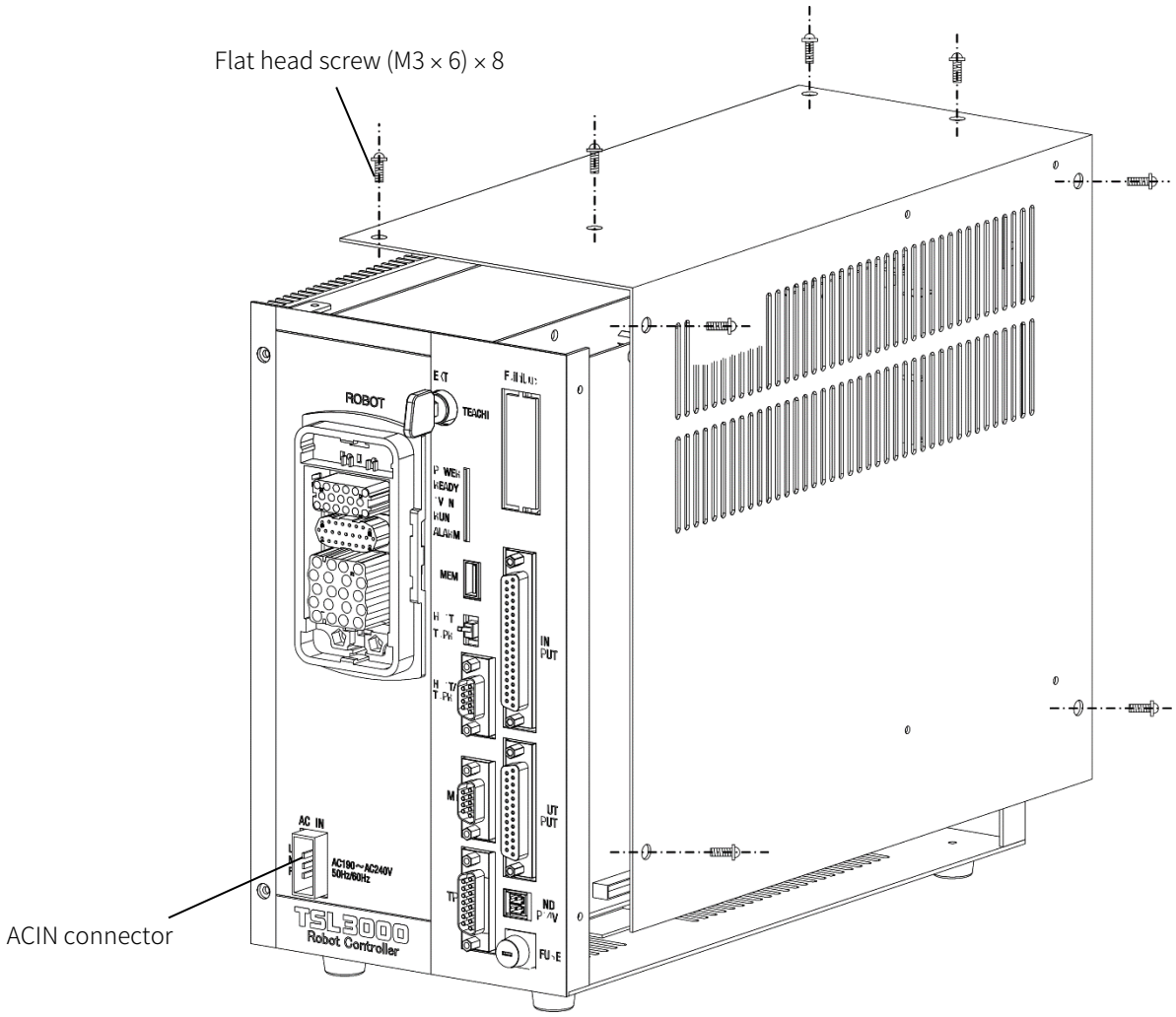



Figure 6.6 Removing the cover

5) Cut the cable ties that hold the battery by nippers, and remove the battery.
(Refer to Figure 6.5 for the battery position.)
6) Remove the battery connector from CN5 of the X8YC board. Immediately connect a new battery connector to CN5. (If the controller is left with battery disconnected for twenty minutes or more, data in the memory will be lost.)
The lithium battery discharges even when it is not used.
Use a lithium battery purchased within three years and has been stored at normal temperature.
7) Fix the battery connected to CN5 with cable ties.
8) Mount the cover with the eight flat head screws (M3 × 6).
9) If the controller is left without battery for twenty minutes or more, connect the controller's power plug to the power supply, and load the programs from the USB flash drive or the computer on which TS Assist (option) is installed.
If the controller detects a battery voltage drop alarm before the five years of lifespan passes, replace the battery immediately.

### 6.3.5 Switching power supply replacement

| DANGER | |
|---|---|
|  **Instruction** | ● Switching power supplies must be installed as described in the instruction manual. PS1 and PS2 switching power supplies have the same mounting pitch. Be careful not to install them wrong way around, or not to use different connectors. Doing so may cause a short circuit or wire breakage, resulting in a fire or an electric shock. |

The lifespan of the switching power supplies (5 VDC, 24 VDC) used for the controller depends on the use condition. Since the aluminum electrolytic capacitor is used, the lifespan will be shorter with the higher load current, longer operating hours, and higher ambient temperatures.
The lifespan of the power supply may extend from 5 years to 10 years depending on the use condition, but we recommend replacing the power supply every five years as preventive maintenance.
This replacement work is difficult to be performed by the customer. Contact our service department.

Replacement Procedure

1) Turn off the breaker for the power supply, and remove the power plug from the power supply.
2) Remove the ACIN connector from the controller.
3) Remove the key from the controller.
4) Using the Phillips screwdriver (#2), remove the flat head screws (M3 × 6 × 8 pcs) fixing the cover, and remove the cover from the unit. (*Mount the cover firmly by tightening the screws after the switching power supply is replaced.)

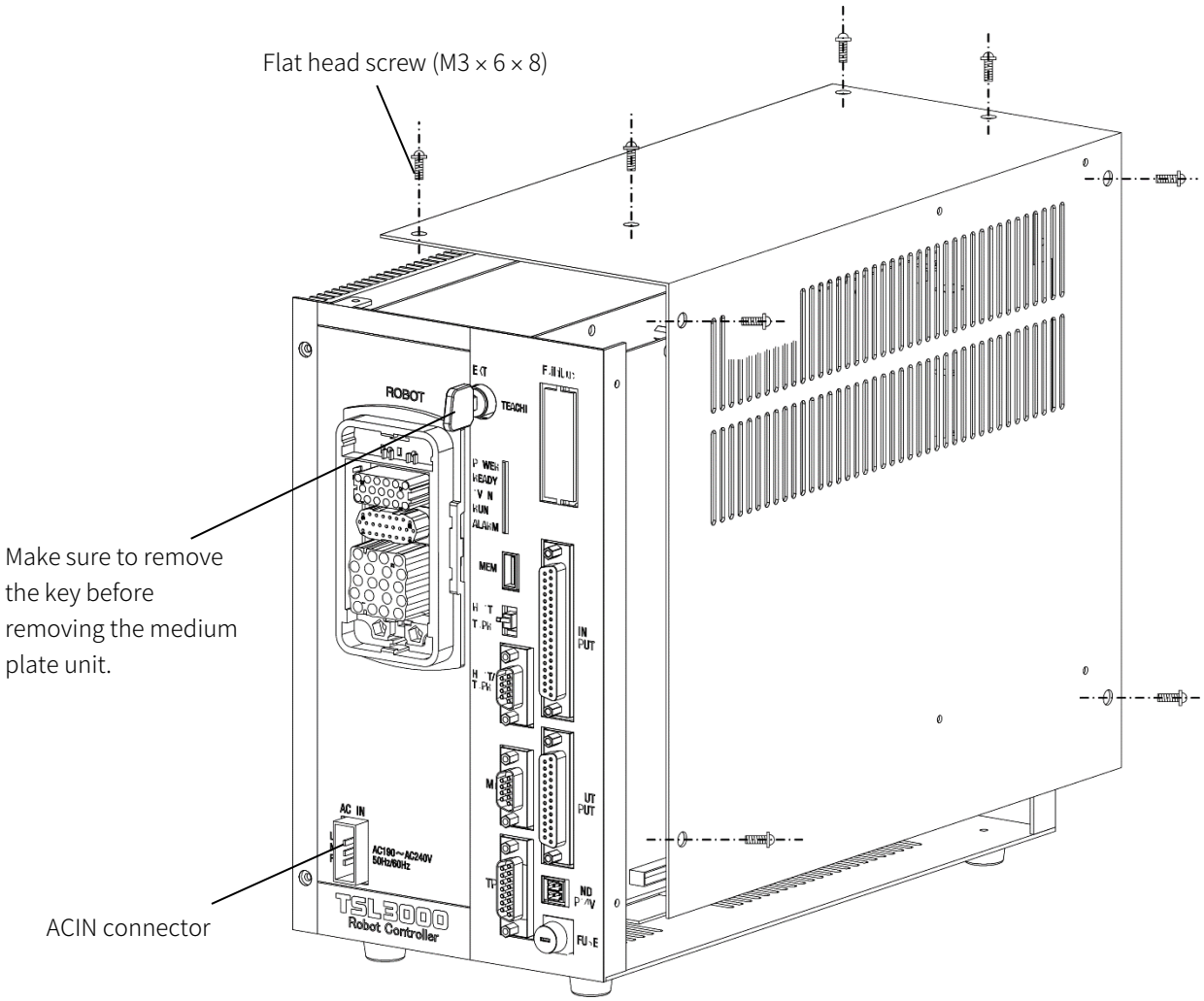



Figure 6.7 Removing the cover

5) Disconnect the DC cable (blue and red), SVIF cable (blue), and encoder cable (blue) from the X8YC (main) board, and the hand I/O cable (blue) and brake cable (blue) from the X8YX (I/O) board.
6) Disconnect the cables (red × 1, blue × 1, black × 2) from PS1 and PS2.

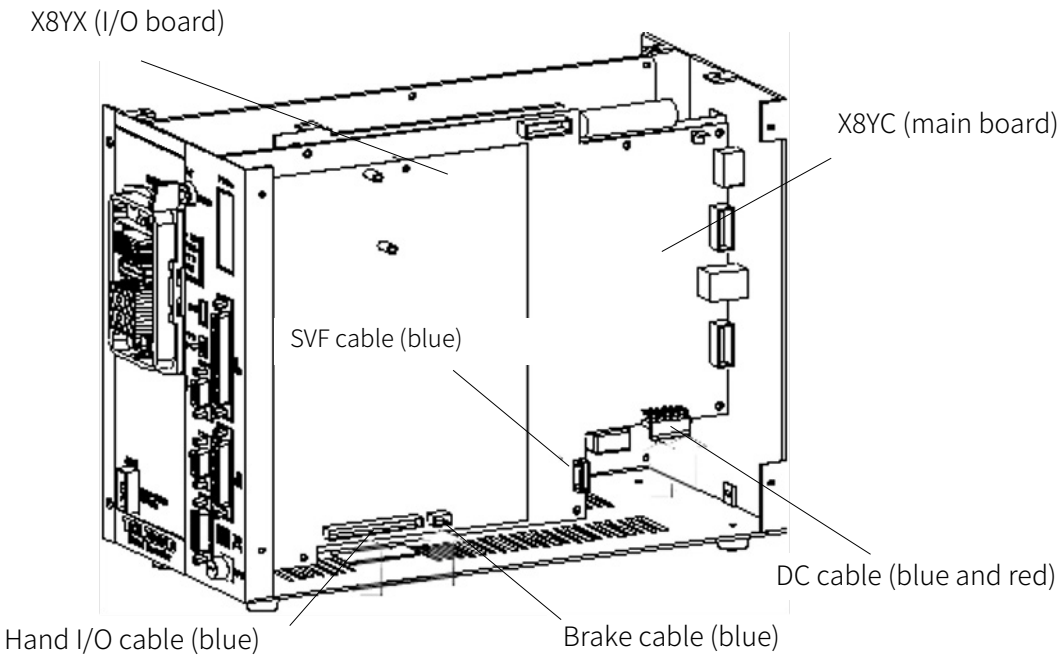



Figure 6.8 Inner side of the controller

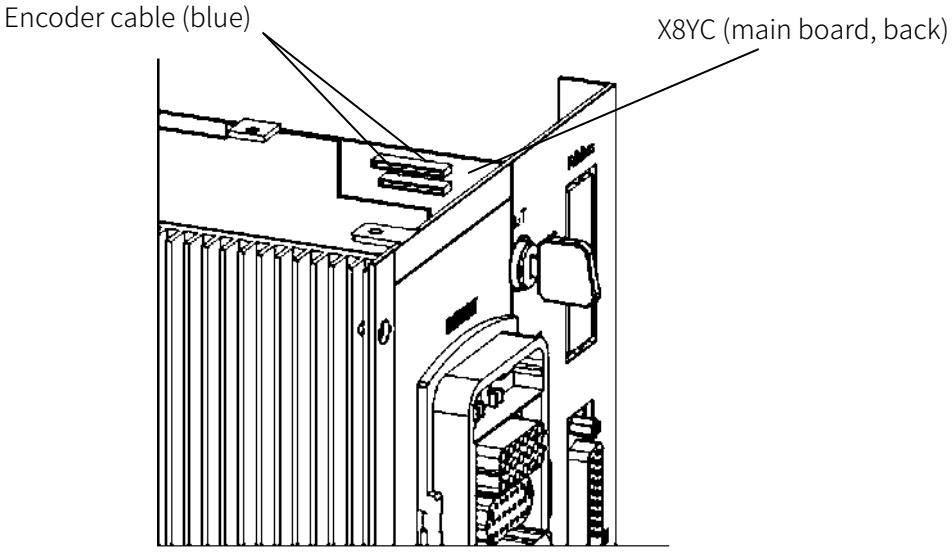



Figure 6.9 Inner front of the controller

7) Remove the flat head screws (M3 × 6 × 1 pcs) that fasten the intermediate plate unit.
Pull out the intermediate plate unit toward the rear side of the controller.
When removing the intermediate plate unit, be sure to remove the key.

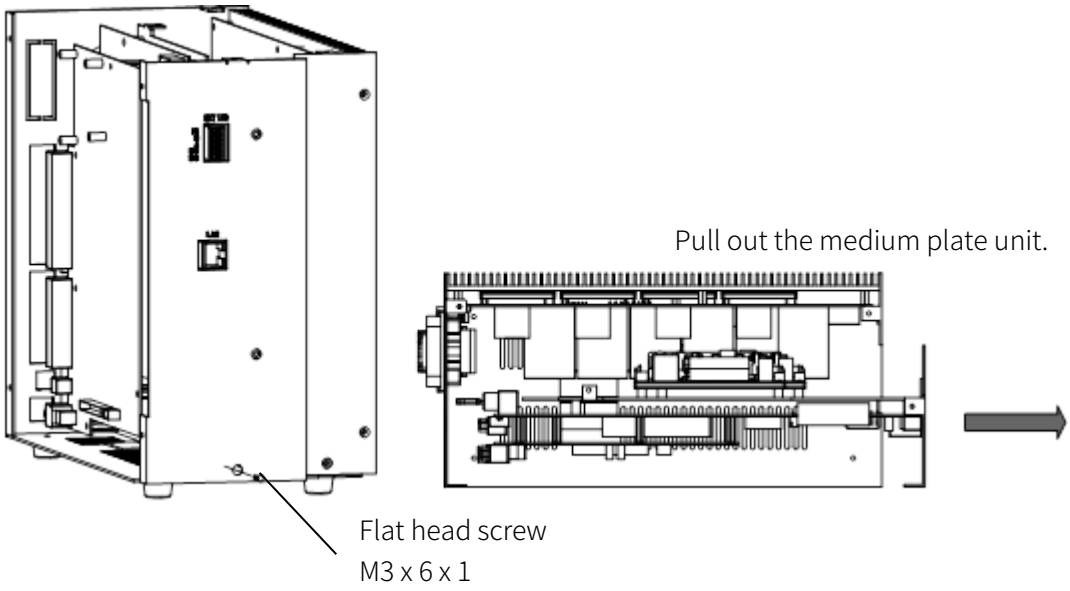



Figure 6.10 Removing the medium plate unit

8) Remove four Sems screws (M3 × 8) from each of the switching power supplies (PS1 and PS2), and remove the switching power supplies from the medium plate unit.

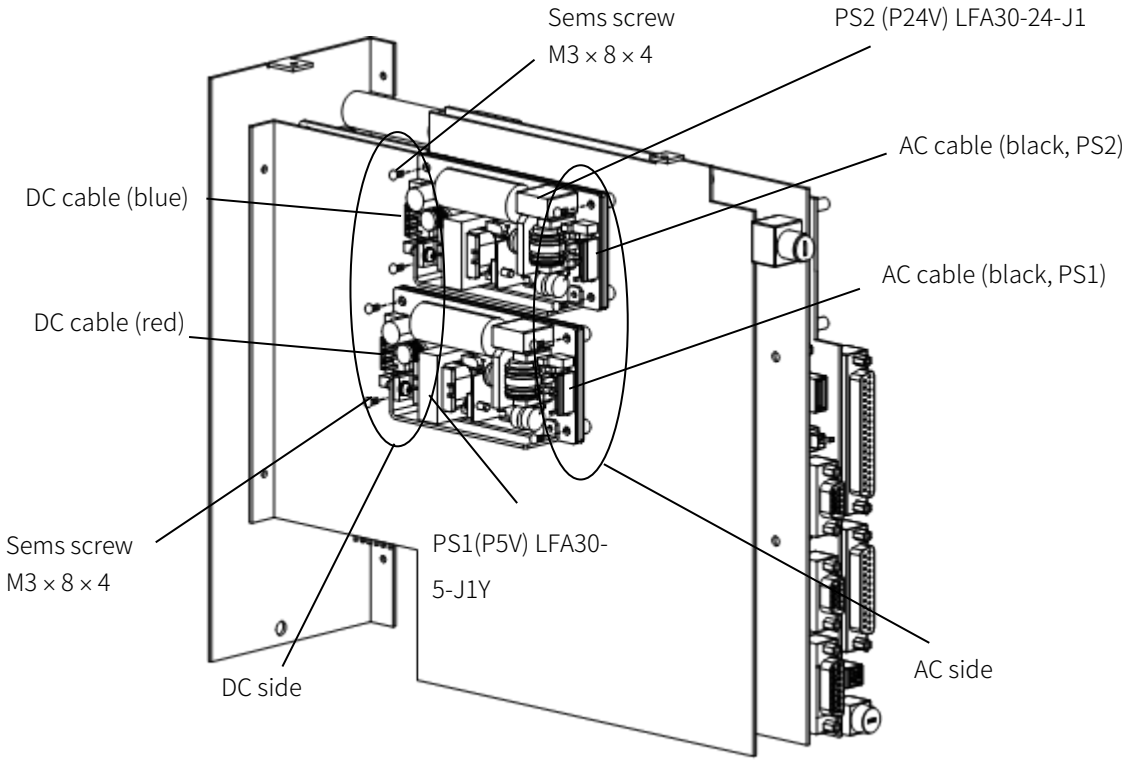



Figure 6.11 Medium plate unit side view

9) Mount each switching power supply with four Sems screws (M3 × 8) to the unit.
   Note: The PS1 and PS2 switching power supplies have the same mounting pitch. Do not set them upside-down or left-right inverted, or do not switch their positions.
10) Connect cables to PS1 and PS2.
   Note: Connect the DC cable (red) to the DC side of the PS1 and the AC cable (black, PS1) to the AC side of PS1, and connect the DC cable (blue) to the DC side of PS2 and the AC cable (black, PS2) to the AC side of PS2.
11) Insert the pulled-out medium plate unit to the controller and mount it with the flat head screw (M3 × 6).
12) Connect the DC cable (blue and red), SVIF cable (blue), and encoder cable (blue) to the X8YC (main) board, and the hand I/O cable (blue) and brake cable (blue) to the X8YX (I/O) board.
13) Mount the cover with the eight flat head screws (M3 × 6).

### 6.3.6 Fuse replacement (X8YX board)

The fuse installed on the front of the controller blows out when the current higher than the specification value flows in the I/O part. If the "I/O Fuse Broken" (8-273) alarm is displayed, the fuse has been blown out. Replace the fuse.

If the fuse of the output module blows out, thoroughly investigate which circuit was in failure, and remove the cause before replacing the fuse.

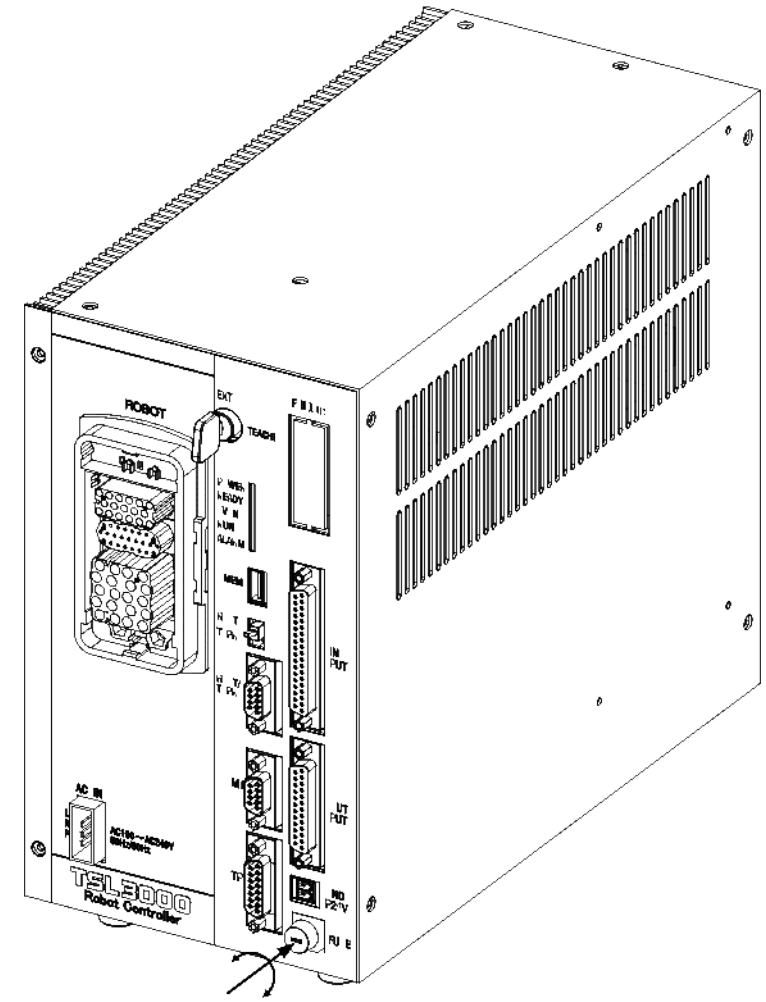



Figure 6.12 Fuse replacement

Table 6.3 Fuse replacement table

| Fuse type | Manufacturer |
|---|---|
| 51NM030H | PICO |

Replacement procedure

1) Disconnect the power plug from the power supply of the controller main unit.
2) Use the flat head screwdriver (#2) to remove the fuse holder in the above figure. (Hold down and turn the holder 90 degrees counterclockwise to unlock it.)
3) Remove the fuse, and attach a new fuse to the fuse holder.
4) Mount the fuse holder. (Hold down and turn the holder 90 degrees clockwise to lock it.)
5) Connect the power plug of the controller main unit to the power supply, and confirm that no error occurs.

### 6.3.7 Output part IC replacement (X8YX board)

A transistor output IC of the X8YX board will be damaged if the current higher than the specification value flows in the output part. In this case, replace the IC. Before replacement, thoroughly investigate which circuit was in failure, and remove the cause.

The following figure shows the output ICs and corresponding output signals.

This replacement work is difficult to be performed by the customer. Contact our service department.

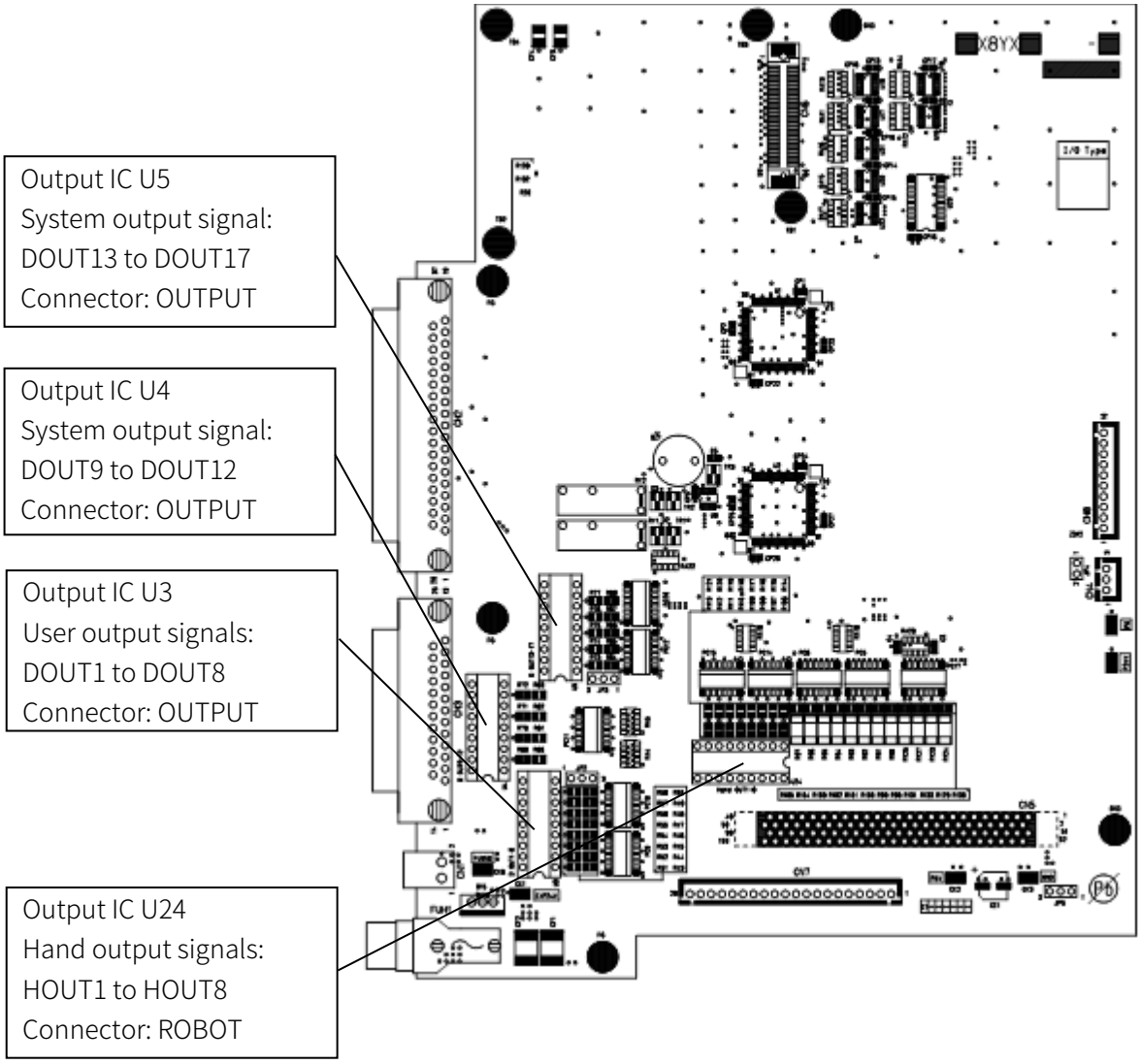



Figure 6.13 I/O board (X8YX)

The transistor output ICs are available in two types. For details, refer to the following table.

Table 6.4 Transistor output IC

| | Output type | Transistor output IC type | Manufacturer |
|---|---|---|---|
| ① | Type-N | ULQ2804A | STMicroelectronics |
| ② | Type-P | M54562WP | MITSUBISHI ELECTRIC |

Replacement procedure

1) Turn off the breaker for the power supply.
2) Remove the ACIN connector from the controller.
3) Remove the eight flat head screws (M3 × 6) fixing the cover, and remove the cover from the unit.
   (* Mount the cover firmly by tightening the screws after replacement.)

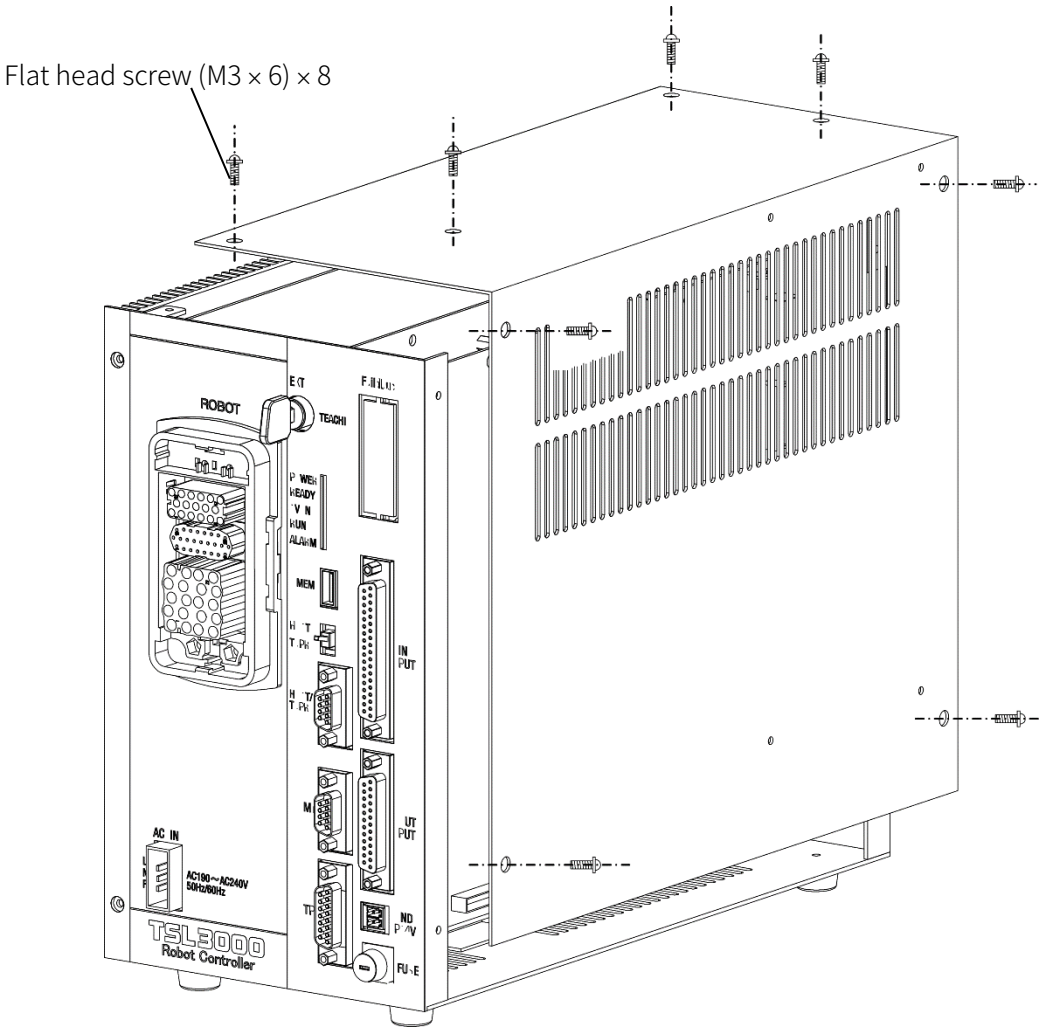



Figure 6.14 Removing the cover

4) Disconnect the hand I/O cable (blue) and brake cable (blue) from the X8YX (I/O) board.
5) Remove the five Sems screws (M3 × 8) fixing the X8YX board, and detach the X8YX board.

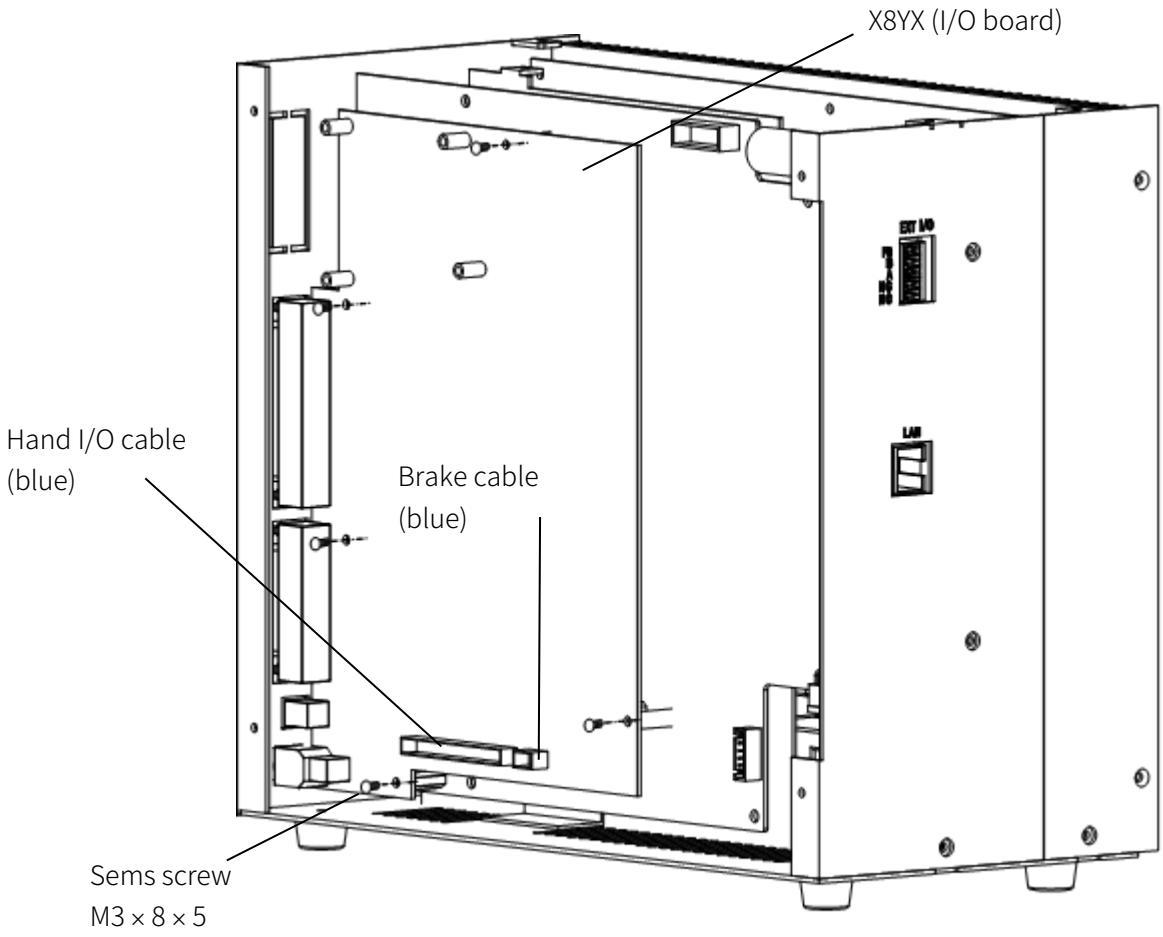



Figure 6.15 Detaching the X8YX board

6) Remove a transistor output IC from the socket as shown in Figure 6.13.
7) Connect a new transistor output IC to the socket.
   Note: Ensure that the model number of the transistor output IC is correct, and connect it in the correct orientation.
8) Mount the X8YX board to the unit with the five Sems screws (M3 × 8).
9) Connect the hand I/O cable (blue) and brake cable (blue) to the X8YX (I/O) board, and mount the cover to the unit with the eight flat head screws (M3 × 6).
10) Connect the power plug of the controller main unit.

# 7 Replacement Parts for Maintenance

## 7.1 Precautions Regarding Replacement Parts

| DANGER | |
|---|---|
|  **Instruction** | ● Be sure to use our specified maintenance and replacement parts. Use of parts other than those specified will result in unexpected accidents.<br>● Except for the battery for the position detector, all parts are of dedicated specifications. To make a purchase or place an order, be sure to contact us. |
|  **Caution/ Warning** | ● For maintenance and replacement work other than fuse replacement, contact our service company. The maintenance/replacement service is available for a fee. We shall not provide any warranty for accidents or damages caused by the work performed by the customer or as a result thereof. |

| NOTICE | |
|---|---|
|  **Instruction** | ● Be sure to use our specified maintenance and replacement parts. Use of parts other than those specified will result in deterioration of the robot's performance and service life, as well as malfunction. |

## 7.2 Replacement Parts List

| No. | Name of product | Model | Unit code | Manufacturer | Use | Remarks |
|---|---|---|---|---|---|---|
| 1 | PS1 (switching power supply) | LFA30F-5-J1Y | | COSEL | 1 | P5V power supply |
| 2 | PS2 (switching power supply) | LFA30F-24-J1 | | | 1 | P24V power supply |
| 3 | Lithium battery | ER6VC4 (ER6C WK27: discontinued) | | Toshiba | 1 | For X8YC board |
| 4 | Fuse | 51NM030H | | PICO | 1 | For X8YX board |
| 5 | Printed board (main control) | X8YCB(TSL3KMAIN) | Y610A90B0 | SHIBAURA MACHINE | 1 | |
| 6 | Servo amplifier module | X8YSB(TSL3KSV) | Y610A9040 | SHIBAURA MACHINE | 1 | For THL500 to THL1200 |
| | | TSL3K-HL4 | Y610A9050 | | 1 | For THL300 and THL400 |
| | | TSL3K-HE4 | Y610A9090 | | 1 | For THE400 |
| 7 | Printed board (I/O board) | X8YXA(TSL3KIO-) | Y610A9020 | SHIBAURA MACHINE | 1 | Type-N |
| | | X8YXB(TSL3KIO+) | Y610A9030 | | 1 | Type-P |
| 8 | Transistor output IC | ULQ2804A | | STMicroelectronics NV | 4 | Type-N |
| | | M54562WP | | MITSUBISHI ELECTRIC | 4 | Type-P |
| 9 | Teaching pendant | TP5100-5 | Y610A43J0 | SHIBAURA MACHINE | 1 | 5 m cable |
| | | TP5100-10 | Y610A43K0 | | 1 | 10 m cable |
| | | TP5100-15 | Y610A43L0 | | 1 | 15 m cable |
| | | TP3000-5 | Y610A43A0 | | 1 | 5 m cable |
| | | TP1000-5 | Y610A2600 | | 1 | Discontinued |
| 10 | System disk | DVDR-KUMI-FHT-MANUAL | Y610D0740 | SHIBAURA MACHINE | 1 | System files and manuals |

## 7.3 List of Accessories

The whole set of accessories can be purchased with the unit code Y610A91B0.

| No. | Name of product | Model | Manufacturer | Use | Remarks |
|---|---|---|---|---|---|
| 1 | I/C jumper connector | M067817 | SHIBAURA MACHINE | 1 | For INPUT |
| 2 | TP jumper connector | M067819 | | 1 | For TP |
| 3 | Connector | XM2S-2511 | OMRON | 2 | Shell |
| 4 | | 03JFAT-SAYGF-I | Japan Solderless Terminal MFG | 1 | For ACIN |
| 5 | | J-FAT-OT | | 1 | ACIN mounting bracket |
| 6 | | XM3A-2521 | OMRON | 2 | For OUTPUT, HAND |
| 7 | | SL-4000-CP-5PGY | SATO PARTS | 1 | For extended I/O |
| 8 | | SL-4000-CP-2PGY | | 1 | For EXT24V |
| 9 | Fuse | 51NM030H | PICO | 1 | EXT24V fuse |
| 10 | Switch | 140007050010 | NIDEC COPAL ELECTRONICS | 1 | For mode key |

# Disposal

This part describes the disposal of the controller.

# 8 Disposal

The controller shall be disposed of according to your company's rules. We do not collect and dispose of your controller.

| DANGER | |
|---|---|
|  **Prohibited** | ● Never incinerate, short circuit, charge, disassemble, or heat batteries. Doing so will result in rupture and fluid leaking. If the leaked fluid gets on your skin or eyes, irritation may occur.<br>• If the leaked fluid gets on your skin, rinse off immediately with running water.<br>• If the leaked fluid gets on your eyes, rinse off immediately with running water and then consult a doctor.<br>• Dispose of batteries according to your company's rules. |

| NOTICE | |
|---|---|
|  **Instruction** | ● Packaging materials that are no longer needed shall be disposed of according to your company's rules. When storing them, be sure to control them so that they are not used for any purpose other than packing the robot. |